# What produces a capturable complexity gap:
# one representative economy per model and group

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**Scope.** Every economy with a protocol run (the committed record in `variants/results/economy_table.csv` plus the October 6–15, 2026 Quest campaign in `bop_runs/results`) and the reduced-form screens. The main tables hold one representative per model per group, built so that the same change is made in all three models; everything else that was run is in the appendix (Section 6). The tables are generated by `variants/campaign_tables.py` from the oracle and estimator files. Every campaign economy ran ten seeds (`gvypk3` twenty; the reduced form five per cell).

**Reading the tables.** Every table has the same columns. $\mathrm{SR}_{\max}$, the *linear ceiling* and the *room* are the population investor's conditional Sharpe ratios over the evaluation window: the linear ceiling is the best Sharpe a model linear in the ranks of the characteristics reaches with the true conditional moments, the room is the best nonlinear ceiling (random features, bins, quadratic) minus that linear ceiling, and *room % of linear* is the room as a percentage of the linear ceiling. That is the theoretical scope for nonlinearity, before any estimation. Sharpe ratios add in quadrature, so the room understates an interaction that sits on a large linear base; *interaction SR* is the nonlinear ceiling's Sharpe orthogonal to the linear one, $\sqrt{\mathrm{nl}^2 - \mathrm{lin}^2}$, the size of the interaction itself, and *DKKM ⊥* is DKKM's Sharpe orthogonal to the best classical method, signed like the gap. *Market share* is the equal-weighted market's share of $\mathrm{SR}_{\max}$. *DKKM*, *FMR* (Fama–MacBeth), *FF* (Fama–French factors), *linear* (the best of rank and level regressions, with and without the market) and *CAPM* (the estimated market line) are each method's out-of-sample Sharpe at the 360-month window. All Sharpe ratios are monthly (multiply by $\sqrt{12}$ for annual), so a gap of 0.01 is 0.035 a year. The *fair gap* is DKKM minus the best of the four classical methods, averaged across seeds, with in parentheses its $t$-statistic (the mean over its standard error across seeds; ten seeds unless stated); "+" counts the seeds in which it is positive. A dagger marks an oracle that predates the evaluation-window ceilings.

## 1 The answer

Three structural asset-pricing models, BGN (Berk, Green and Naik 1999), GS21 (Gomes and Schmid 2021) and KP14 (Kogan and Papanikolaou 2014), are each simulated as a panel of 500 firms over 500 months, and every simulated economy is scored the same way. DKKM's random-feature portfolio is run against the best classical linear method (Fama–MacBeth, Fama–French, a linear model in the ranks of the characteristics with or without the market, and the market line), out of sample at a 360-month window, and the *fair gap* is DKKM's Sharpe ratio minus the best of these. Behind it sits the *room*: on the true conditional moments, the Sharpe ratio of the best nonlinear portfolio of the characteristics minus that of the best linear one. Room is necessary for a gap and not sufficient. Every number is a monthly Sharpe ratio averaged over ten seeds, with the $t$-statistic of the mean across seeds; a gap of +0.029 a month is 0.1 a year, the bar for an economy to count.

Each model starts from its published baseline, and each section changes one thing about it, the same change in all three models.

- A priced factor whose only priced event is a crash (Section 1): no gap. Room is 3–12% of the linear ceiling.
- A price of risk that rises with the square of the aggregate state (Section 2): no gap. Room 0–14%.
- A priced cash-flow factor to which every firm is exposed through one of its own states (Section 3a): no gap. Room 5–10%: a monotone exposure, however nonlinear, is linear in the rank of the characteristic that reveals it.
- An exposure that is the product of two firm-level factors, one of them a sign the characteristics do not carry directly (Section 3b): a gap in all three models, +0.071 in BGN, +0.059 in GS21, +0.091 in KP14.
- An exposure that rises and falls in a characteristic (Section 3c): a gap in all three, +0.119 in BGN and +0.212 in KP14 (a hump across size), +0.080 in GS21 (quadratic in productivity).
- More priced factors (Section 4) help only when each factor's loading is itself a product of partly hidden states or the square of a characteristic; factors loading linearly on observed characteristics give nothing at any number of factors.

The six economies, one of each kind per model, and the reading we give them:

- **BGN interaction** (`bvychs35`, +0.071, $t$ 5.8): six types of firm, the winners and the losers of the priced shock (read: industries), with the exposure scaled by how much of the firm's value is in operating assets rather than growth options, which bear little of the shock.
- **GS21 interaction** (`gvysz2`, +0.059, $t$ 11.5): the same winners and losers, with the exposure scaled by the firm's own productivity: a reallocation shock, in which the productive firms of a losing industry gain at the expense of the weak.
- **KP14 interaction** (`vygelbcs`, +0.091, $t$ 12.3): winners and losers, with the exposure scaled by the firm's profitability relative to normal: a dispersion shock, in which an exposed industry's strong firms pull further ahead and its weak firms fall further behind.
- **BGN and KP14 non-monotone** (`bvyhx4`, +0.119, $t$ 10.9; `vyhx3`, +0.212, $t$ 12.5): five size classes, the smallest and the largest exposed to the shock with one sign and the mid-sized with the other, so that exposure rises and falls with size.
- **GS21 non-monotone** (`gvyzzh10`, +0.080, $t$ 4.6): a firm's exposure is quadratic in its own productivity; firms of middling productivity bear the shock, the most productive have the margin to absorb it and the least productive have already shed the capacity that carries it.

All ten seeds are positive in each of the six. Section 6 lists every other economy that was run.

Why these and not the others. Every factor model is firm exposure times aggregate shock, and a linear model in the ranks of the characteristics fits any additive function of them. A gap needs an exposure with no additive part for the characteristics to carry: the product of two factors that are each mean zero across firms, or a function that rises and then falls in one characteristic. With one priced factor this is all or nothing. A linear portfolio with any covariance with the exposure mimics the factor, diversifies the rest and takes its whole Sharpe ratio, so the interaction economies need the sign hidden, and the non-monotone economies need the premium to be non-monotone in whatever the characteristics reveal about the firm. Where the premium is monotone in something revealed, as for GS21's size ladder, for leverage fixed by type, or for a one-signed U in size (Section 6), the linear model holds it however curved the exposure is.

The baselines every section is read against:

Table 1: Baselines. The three models as published, one type each: the floor every group is read against.

| model | economy | $SR_{\max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bgnbase` – baseline, no priced state beyond the rate | 0.301 | 54% | 0.250 | +0.027 | 11 | 0.12 | 0.238 | 0.213 | 0.208 | 0.238 | 0.162 | -0.0001 (t -0.0) | +0.00 | 6/10 |
| GS21 | `gsbase` – baseline | 0.293 | 95% | 0.291 | +0.001 | 0 | 0.03 | 0.283 | 0.260 | 0.258 | 0.286 | 0.278 | -0.0028 (t -4.3) | -0.03 | 2/10 |
| KP14 | `kpbase` – baseline, one type, no priced state | 0.229 | 35% | 0.221 | +0.004 | 2 | 0.04 | 0.208 | 0.193 | 0.195 | 0.209 | 0.078 | -0.0010 (t -0.6) | -0.00 | 6/10 |

**Notation.** BGN (Berk, Green and Naik 1999): a firm is a portfolio of live projects; project $j$ has systematic risk $\beta_j$, drawn from an exponential distribution, and lognormal cash flows $C_j$; it survives each month with probability $\pi$; a new project arrives each month and is taken if its NPV is positive; the pricing kernel prices the market shock at $\sigma_z$ and the Vasicek short rate $r_t$. GS21 (Gomes and Schmid 2021): output is $e^{x_t+z_{it}}k_{it}$ with aggregate productivity $x_t$ and idiosyncratic productivity $z_{it}$ both AR(1); firms choose the coupon $b_{it}$, invest lumpily at a random cost, and may default; the kernel prices $x$-shocks at a constant $\gamma_x = 0.5$. KP14 (Kogan and Papanikolaou 2014): firm value is assets in place plus growth options; project cash flows are $x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, new projects arrive at the firm-specific rate $\lambda_{it}$, and the kernel prices disembodied productivity $x$ at $\gamma_x$ and embodied technology $z$ at $\gamma_z$.

**The added priced state $y$.** Every economy with a $y$ adds one stationary state with unit stationary variance and mean reversion $\kappa_y = 0.35$ per year, $y_{t+1} = \rho y_t + \sqrt{1-\rho^2}\,\eta_{t+1}$ with $\rho = e^{-\kappa_y/12}$ (an Ornstein–Uhlenbeck process in KP14, a Tauchen chain in GS21 and in BGN's chain economies). It is priced: the kernel is multiplied by $e^{-\lambda_y \eta_{t+1} - \lambda_y^2/2}$ with $\lambda_y = \gamma_v \sqrt{1/12}$, and $\gamma_v = 2.5$ unless stated. "Types" are equal shares of firms differing only in their exposure $b_f$ to $y$. "Level-compensated" means type $f$'s cash flows are also scaled by a constant $\theta_f = (V_0/V_f)^c$, the ratio of a unit project's value for a reference type and type $f$ at $y = 0$, so that value at $y = 0$ does not reveal the type .

**The baselines.**

- `bgnbase`: BGN as above, one type. Expected returns come from the value-weighted $\beta_j$ of assets in place and from the growth option's exposure to $r_t$.
- `gsbase`: GS21 as above, Table I calibration, one type.
- `kpbase`: KP14 as above, one type, no $y$.

## 2 A different distribution of shocks: a crash-only factor

A cash-flow factor $y$ (unit variance, half-life two years) multiplies every firm's cash flows as $e^{by}$, with one exposure $b$ per model: 0.035 in BGN's project cash flows, $0.10\,\tilde{z}$ in GS21's output (the product economy's exposure without the coupon), 0.07 in KP14's project cash flows. Its Gaussian shocks carry no price. Its only priced event is a crash: each month, with probability 1%, $y$ jumps to $-3.5$ standard deviations, and the kernel weights that transition by $e^2$ (about 7% a month risk-neutral); $y$ then recovers at its ordinary mean reversion. Compared with the one-type economy of Section 3a, the only change is that the Gaussian price is replaced by the crash.

Table 2: Group 1. A different distribution of shocks: a crash-only factor. Every firm's cash flows load on a factor y whose only priced shock is a rare crash (1% a month to $-3.5$ sd, kernel weight $e^2$).

| model | economy | $\mathrm{SR}_{\mathrm{max}}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvy1j` – crash-only factor: $e^{0.035y}$, the same crash | 0.324 | 50% | 0.278 | +0.033 | 12 | 0.14 | 0.265 | 0.239 | 0.235 | 0.262 | 0.160 | +0.0003 (t 0.1) | +0.01 | 6/10 |
| GS21 | `gvyz10j` – crash-only factor on $0.10\,\tilde{z}\,y$: y unpriced except the 1%/month jump to $-3.5$ (weight $e^2$) | 0.386 | 82% | 0.374 | +0.011 | 3 | 0.09 | 0.358 | 0.340 | 0.333 | 0.362 | 0.316 | -0.0046 (t -3.5) | -0.05 | 1/10 |
| KP14 | `vy1j` – crash-only factor: $e^{0.07y}$, y unpriced except a 1%/month jump to $-3.5$ (weight $e^2$) | 0.336 | 21% | 0.277 | +0.012 | 4 | 0.08 | 0.259 | 0.241 | 0.249 | 0.265 | 0.066 | -0.0057 (t -2.3) | -0.04 | 2/10 |

**Verdict.** A crash changes the distribution of a shock every firm bears in the same proportion. The cross-section of premia it creates is a sort on the one firm state that scales the exposure, so the linear ceiling takes nearly all of it: room is 12% of the linear ceiling in BGN, 3% in GS21 and 4% in KP14, and the fair gap is zero or negative in all three (+0.000, $-0.005$, $-0.006$). Adding the same crash to the interaction economies instead (Appendix A1) leaves their room and gap at the control's in BGN and GS21 and halves KP14's.

**What changes.** The 3a economy with the Gaussian price of $y$ set to zero ($\gamma_v = 0$) and a crash.

- `bvy1j`, `gvyz10j`, `vy1j`: `bvy1ch`, `gvyz10` and `vy1` with $\gamma_v = 0$ and, each month with probability 1% whatever $y_t$, a jump of $y_{t+1}$ to $-3.5$ (the lowest node), weighted $e^2$ by the kernel (a risk-neutral probability of about 7% a month before renormalization). In KP14 the jump is a Poisson event at intensity 0.12 a year under $P$ and $0.12\,e^2$ under $Q$.

## 3 A different price of risk: quadratic in the model's own state

The price of each model's own aggregate risk is multiplied by $1 + 0.25\,s^2$, capped at 3, where $s$ is a standardized aggregate state: GS21's productivity $x$, and for KP14, whose own states are random walks, an unpriced stationary state $y$ that no firm is exposed to. The multiplier never falls below 1 because KP14's growth-option discount turns negative once the price of $x$-risk drops below about 0.82 of its value. In BGN the price of the market shock $\sigma_z$ is multiplied by the same function of the standardized rate, binned to seven equiprobable levels of the rate (1.00, 1.03, 1.17 and 1.68 from the middle outward) so that project values keep a closed form; the date-$k$ cash flow is priced at the bin of the rate one period before it pays.

Table 3: Group 2. A quadratic price of the model's own risk. The price of the baseline's aggregate risk becomes $1 + 0.25s^2$ times its value, capped at 3, $s$ a standardized aggregate state.

| model | economy | $\mathrm{SR}_{\mathrm{max}}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bq25` – $\sigma_z \times \min(1 + 0.25\,s^2, 3)$, s the standardized rate, 7 bins | 0.411 | 36% | 0.346 | +0.049 | 15 | 0.18 | 0.339 | 0.303 | 0.293 | 0.335 | 0.149 | +0.0034 (t 0.7) | +0.00 | 4/10 |
| GS21 | `gq25` – $\gamma_x \times \min(1 + 0.25\,\tilde{x}^2, 3)$ | 0.345 | 95% | 0.343 | +0.002 | 0 | 0.03 | 0.334 | 0.316 | 0.309 | 0.337 | 0.326 | -0.0036 (t -5.5) | -0.05 | 0/10 |
| KP14 | `kpq` – $\gamma_x \times \min(1 + 0.25\,y^2, 3)$, y unpriced | 0.260 | 53% | 0.251 | +0.003 | 1 | 0.04 | 0.240 | 0.226 | 0.225 | 0.242 | 0.137 | -0.0021 (t -1.5) | -0.02 | 4/10 |

**Verdict.** A price of risk that moves with the aggregate state rotates the premium across states and leaves it linear in the ranks within one. The quadratic price gives room under 1% of the linear ceiling in GS21 and KP14, and 14% in BGN, where the rate is itself a priced state so a price that rises with its distance from the mean rotates the premium with it; the fair gap is zero in all three (+0.003, $-0.004$, $-0.002$). The regime prices tried earlier (appendix) do the same, and BGN's fast regime gave the group's one small positive number (+0.008), which a ranks-times-state linear benchmark takes.

**What changes.**

- `bq25`: `bgnbase` with $\sigma_z \to \sigma_z\, g(r_t)$, $g(r) = \min(1 + 0.25\,((r - \bar{r})/\mathrm{sd}_r)^2, 3)$ binned to seven equiprobable bins of the stationary rate ($g = 1.00$, 1.03, 1.17, 1.68 from the middle outward); the date-$k$ cash flow is priced at the bin of $r_{t+k-1}$, through seven bases $D_j(r)$ in place of the regime's two.
- `gq25`: `gsbase` with $\gamma_x \to \gamma_x(1 + 0.25\,\tilde{x}^2)$ capped at 1.5, $\tilde{x} = x/\mathrm{sd}(x)$.

- `kpq`: `kpbase` with $\gamma_x \to \gamma_x \min(1 + 0.25\, y^2, 3)$, $y$ the unpriced stationary state of `kpgam` ($\kappa = 0.35$, unit variance); no firm is exposed to $y$.

# 4 Different exposures to aggregate risk

## 4.1 Exposure that a single characteristic captures

The priced state $y$ is added with one type: every firm's cash flows load on it with the same exponent (the group 1 exposures, now with $y$'s Gaussian shocks priced at 2.5). A firm's exposure to $y$ then varies across firms only through one firm state: the mix of assets in place and growth option in BGN, productivity in GS21 (the product economy's exposure without the coupon), the growth-option share in KP14.

Table 4: Group 3a. Exposure captured by a single characteristic. A priced state y with one type: a firm's exposure varies only through one firm state (project mix, operating leverage, growth-option share).

| model | economy | $\mathrm{SR}_{\max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvy1ch` – one type, project cash flows $\times e^{0.035y}$ | 0.336 | 54% | 0.297 | +0.029 | 10 | 0.13 | 0.283 | 0.252 | 0.255 | 0.281 | 0.181 | +0.0017 (t 0.8) | +0.02 | 7/10 |
| GS21 | `gvyz10` – one type, output $\times e^{0.10\,\tilde{z}\,y}$: the product economy without the coupon | 0.854 | 50% | 0.800 | +0.043 | 5 | 0.27 | 0.793 | 0.805 | 0.742 | 0.808 | 0.421 | -0.0164 (t -6.5) | -0.15 | 1/10 |
| KP14 | `vy1` – one type, cash flows $\times e^{0.07y}$ | 0.466 | 12% | 0.369 | +0.019 | 5 | 0.12 | 0.355 | 0.335 | 0.340 | 0.360 | 0.047 | -0.0050 (t -2.9) | -0.05 | 1/10 |

**Verdict.** A monotone function of one firm state, however nonlinear, is linear in the rank of the characteristic that reveals it. Room is 10% of the linear ceiling in BGN and 5% in GS21 and KP14, and the fair gap is zero or negative in all three. The same holds for every other single-characteristic economy tried (Section 6), including ones with far more room: room is necessary, not sufficient.

**BGN.** *Setup.* The baseline BGN plus the priced state $y$, a 21-node chain with mean reversion 0.35 a year and unit variance, its Gaussian shocks priced at $\gamma_v = 2.5$. Every live project's cash flow is multiplied by $e^{0.035\, y_t}$, and growth options are valued on the same $y$. One type: one firm's exposure differs from another's only through the share of its value in projects, which bear the shock, rather than growth options, which bear less of it. *Results.* Linear ceiling 0.28–0.33 across seeds, room +0.02 to +0.06 (10% of the ceiling). DKKM 0.283 against the best linear method's 0.282: fair gap +0.002 ($t$ 0.8, seven of ten). *Story.* A cash-flow shock every firm bears in proportion to its operating assets; book-to-market reads the share of assets in place directly, so a linear sort holds the premium.

**GS21.** *Setup.* The baseline GS21 plus $y$ (9-node chain, same mean reversion and price). Output is $e^{x_t + z_{it} + 0.10\, \tilde{z}_{it} y_t} k_{it}$, $\tilde{z}$ the firm's standardized idiosyncratic productivity, so the loading on $y$ is $0.10\, \tilde{z}$. One type. *Results.* Linear ceiling 0.47–0.99, room +0.03 to +0.06 (5%). DKKM 0.793 against 0.810: fair gap −0.017 ($t$ −5.4, one of ten). *Story.* The shock scales with productivity; profitability reads productivity, so the premium is a sort on ROE.

**KP14.** *Setup.* The baseline KP14 plus $y$ (an OU state, same mean reversion and price). Each project's cash flow is $e^{0.07\, y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, the growth option valued on the same $y$. One type; exposure differs only through the growth-option share. *Results.* Linear ceiling 0.35–0.39, room +0.01 to +0.02 (5%). DKKM 0.355 against 0.360: fair gap −0.005 ($t$ −2.9, one of ten). *Story.* As BGN: operating assets bear the shock, growth options do not, and the growth-option share is one number per firm that size and book-to-market read.

## 4.2 Exposure as an interaction of several characteristics

Start from the one-type economy of Section 3a and give the exposure a sign that varies across firms. Firms come in six types of equal share with loadings on $y$ of $\pm b_1, \pm b_2, \pm b_3$: the two signs are the winners and the losers of the priced shock, an industry attribute that the five characteristics do not include, and the cross-sectional mean of the loading is zero. Each type is level-compensated at $y = 0$, so that value does not reveal the sign, and the types sit on a ladder of cash-flow levels, so that size reveals it in part. The type's loading multiplies a firm-level quantity, and this is where the models differ: in BGN the share of the firm's value in assets in place, which the model generates; in KP14 the firm's profitability relative to normal, $\varepsilon - 1$, written into the cash flow; in GS21 the firm's standardized productivity $\tilde{z}$, written into the production function. The exposure is then type sign × firm state, a product of two things that different characteristics reveal. The one-sided versions, the levers tried on the way and the leverage version of GS21's firm state are in Section 6.

Table 5: Group 3b. Exposure as an interaction of characteristics. One construction in every model: a mirrored set of types, the winners and losers of the priced shock, level-compensated at $y = 0$ and revealed only in part through size, times a firm-level amplifier – the assets-in-place share in BGN (bvychs35), the profitability deviation in KP14 (vygelbcs, bounded exposure, price 4.0, $\sigma_\epsilon$ 0.3), the productivity in GS21 (gvysz2).

| model | economy | $\mathrm{SR}_{\max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvychs35` – bvychs at bv_comp 3.5 | 0.508 | 31% | 0.294 | +0.189 | 67 | 0.38 | 0.403 | 0.315 | 0.309 | 0.303 | 0.156 | +0.0712 (t 5.8) | +0.22 | 10/10 |
| GS21 | `gvysz2` – type sign × productivity $\tilde{z}$: mirrored types ±0.15 / 0.3 / 0.5, gs_level ladder 0.5..2.0, no compensation | 0.847 | 17% | 0.603 | +0.208 | 37 | 0.54 | 0.678 | 0.536 | 0.488 | 0.619 | 0.151 | +0.0590 (t 11.5) | +0.27 | 10/10 |
| KP14 | `vygelbcs` – bounded exposure (cap 2.5), mirrored types ±0.3 / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_v$ 4.0, $\sigma_\epsilon$ 0.3 | 0.858 | 7% | 0.297 | +0.256 | 90 | 0.46 | 0.388 | 0.251 | 0.288 | 0.282 | 0.052 | +0.0913 (t 12.3) | +0.25 | 10/10 |

**Why a product has room.** A linear model in the ranks fits additive functions of the characteristics. The product of two factors each mean zero across firms has no additive part, so no linear portfolio is long the exposure, while random features span products. The sign must be hidden: one-sided versions, in which every type loads with the same sign and the firm state alone switches sign, are a sort on that state's characteristic with a type-dependent slope, room 10–27%, no gap (Section 6). The $y = 0$ compensation keeps value from revealing the sign; how visible the type is through size does not matter for room.

**BGN.** *Setup.* The 3a economy with six types of equal share, loadings $b_f \in \{-0.07, -0.035, -0.01, 0.01, 0.035, 0.07\}$ on $y$ in every project cash flow ($C_j e^{b_f y_t}$), each type's cash-flow level multiplied by $\theta_f = (V_0/V_f)^{3.5}$, $V_f$ the value of a unit project of type $f$ at $y = 0$; the exponent 3.5 over-compensates, so size reveals the type in part. The firm's exposure is $b_f$ times its model-generated share of value in assets in place. *Results.* Linear ceiling 0.22–0.38, room +0.13 to +0.23 (67%). DKKM 0.403 against the best linear method's 0.332: fair gap +0.071 ($t$ 5.8, ten of ten); exponents 2.5 / 3.0 / 3.5 / 4.0 give +0.048 / +0.063 / +0.071 / +0.059. *Story.* Industries on opposite sides of a priced shock; within an industry the firm whose value sits in operating assets moves with it and the firm whose value sits in growth options does not. Book-to-market reads the second factor and size part of the first; no single characteristic reads the product.

**KP14.** *Setup.* The 3a economy with the exposure moved onto the profitability deviation: a project's cash flow is $x_t K_j^\alpha \theta_f u_{it} \left[1 + (\varepsilon_{it} - 1)\, h_{b_f}(y_t)\right]$, $\varepsilon$ the firm's profitability (mean 1) and $h_b(y) = c/(1 + (c - 1)e^{-cby/(c-1)})$ a bounded function with slope $b$ at $y = 0$ and values in $(0, c)$, $c = 2.5$ (the exponential would make project values negative for $b \leq -0.2$). Six types $b_f \in \pm\{0.3, 0.7, 1.2\}$, each type's level $\theta_f$ multiplied by 0.5 / 0.65 / 0.85 / 1.15 / 1.5 / 2.0; price of $y$ risk 4.0; $\sigma_\varepsilon = 0.3$ (baseline 0.2). The exposure is $b_f(\varepsilon - 1)h'(y)$: type sign times profitability relative to normal. *Results.* Linear ceiling 0.21–0.43, room +0.15 to +0.30 (58–119%). DKKM 0.388 against 0.297: fair gap +0.091 ($t$ 12.3, ten of ten; +0.041 to +0.131). *Story.* A dispersion shock: in an exposed industry, firms above normal profitability pull further ahead and firms below fall further behind. ROE reads the magnitude, size part of the sign, no single characteristic both.

**GS21.** *Setup.* The 3a economy with six types whose loading multiplies the firm's standardized productivity: output $e^{x_t + z_{it} + b_f \tilde{z}_{it} y_t} k_{it}$, $b_f \in \pm\{0.15, 0.3, 0.5\}$, each type's production level multiplied by 0.5 / 0.65 / 0.85 / 1.15 / 1.5 / 2.0, no compensation (the exposure is mean zero across firms), price 2.5.

*Results.* Linear ceiling 0.32–0.73, room +0.18 to +0.27 (22–66%). DKKM 0.678 against 0.619: fair gap +0.059 ($t$ 11.5, ten of ten); the smaller ladder $\pm\{0.1, 0.2, 0.35\}$ gives +0.053 ($t$ 6.1). The type premium across the ladder, −0.30, +0.01, +0.11, +0.51, +0.42, +0.24% a month, is non-monotone in $b_f$. *Story.* A reallocation shock: in a losing industry the productive firms gain share from the weak. ROE reads productivity and size part of the industry; the product is read by neither.

GS21's product without types, productivity × coupon (`gvyp15`, +0.078), is the same mechanism without the industry sign. Leverage as the firm state does not work as a type: with leverage fixed by type the exposure is a function of a few perfectly observed groups that the linear model prices directly (Section 6).

### 4.3 Exposure that is non-monotone in a characteristic

A monotone exposure in one firm state is linear in that state's rank; an exposure that rises and then falls in a characteristic is not, and a linear model fits a line through the hump or the U. The construction, the same in every model that allows it: five equal types on a mild visible level ladder (the cash-flow or production level × 0.8, 0.9, 1.0, 1.1, 1.2), each level-compensated at $y = 0$ (bv_comp 1.0), the loading hump-shaped, $b = -c, 0, +c, 0, -c$, or U-shaped, $+c$ at the ends, price 2.5. The hump's story is the thinner of the two shapes, the tails of the size distribution on one side of the shock and the middle on the other; the U, with the plainer story (small firms exposed through financing constraints, large firms through operating leverage), reaches the bar only in BGN and only just, and is in Section 6 with the leverage U.

Table 6: Group 3c. Exposure non-monotone in a characteristic. A hump-shaped loading ($b = -c, 0, +c, 0, -c$) across five equal types on a mild visible level ladder (0.8 to 1.2), level-compensated at $y = 0$, in BGN (bvyhx4, $c = 0.28$) and KP14 (vyhx3, $c = 0.21$); GS21, where the ladder is priced linearly, carries the exposure quadratic in the firm's own productivity (gvyzzh10).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.28, +0.00, +0.28, +0.00, -0.28 | 0.642 | 25% | 0.357 | +0.260 | 73 | 0.50 | 0.476 | 0.341 | 0.293 | 0.344 | 0.159 | +0.1186 (t 10.9) | +0.31 | 10/10 |
| KP14 | `vyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.632 | 7% | 0.207 | +0.377 | 197 | 0.54 | 0.431 | 0.217 | 0.136 | 0.178 | 0.031 | +0.2123 (t 12.5) | +0.37 | 10/10 |
| GS21 | `gvyzzh10` – hump-shaped exposure in the firm's own productivity: $-0.1\,(\tilde{z}^2 - 1)$, one type | 0.639 | 25% | 0.364 | +0.250 | 70 | 0.49 | 0.480 | 0.325 | 0.255 | 0.400 | 0.152 | +0.0795 (t 4.6) | +0.25 | 10/10 |

**BGN.** *Setup.* The 3a economy with five types of equal share whose cash-flow levels are multiplied by 0.8, 0.9, 1.0, 1.1, 1.2 and whose loadings on $y$ are $-0.28, 0, +0.28, 0, -0.28$: the smallest and largest firms load one way, the mid-sized firm the other, the two in between not at all; exact compensation at $y = 0$, so only size reveals the loading; price 2.5. *Results.* Room 59–77% of the linear ceiling. DKKM 0.476 against 0.357: fair gap +0.119 ($t$ 10.9, ten of ten; +0.083 to +0.188). The type premium across the ladder is 0.12, 1.69, 4.76, 1.43, 0.09% a month; the ladder rises with the loading (room 9–13% at ±0.14, +0.035 at ±0.21). *Story.* Exposure that rises and falls with size: the smallest and largest firms on one side of the shock, the mid-sized on the other. The reading is thin and the econometric point is not: a sort on size cannot hold a premium that peaks in the middle. The one-signed U (small firms exposed through financing constraints, large firms through operating leverage) has the plainer story and reaches the bar at a loading of 0.35 (+0.0295, $t$ 3.1, nine of ten; Section 6).

**KP14.** *Setup.* The same five types in KP14: cash-flow levels 0.8–1.2, loadings $-0.21, 0, +0.21, 0, -0.21$ on every project cash flow, exact compensation, price 2.5. *Results.* Linear ceiling 0.21–0.23, room +0.35 to +0.38 (154–197%). DKKM 0.431 against 0.218: fair gap +0.212 ($t$ 12.5, ten of ten; +0.134 to +0.306); at ±0.28, +0.166 ($t$ 10.6). The type premium across the ladder is 0.36, 0.86, 0.68, 0.88, 0.35% a month: the unexposed middle types earn the most, because the market beta rises with $b_f$ (0.73 to 1.36) while the $y$ premium falls, so the premium is non-monotone in the exposure itself and no characteristic that reveals the type can rank it. *Story.* As BGN.

**GS21.** *Setup.* The 3a economy with one type whose loading on $y$ is $-0.1\,(\tilde{z}^2 - 1)$: output

$e^{x_t+z_{it}-0.1(\tilde{z}_{it}^2-1)y_t}k_{it}$, a loading mean zero across firms and hump-shaped in productivity; price 2.5. The size ladder itself gives nothing in GS21: a type's production level moves its capital, value and profitability together, so book-to-market and ROE rank the exposure (`gvyhx3`: room 9–31%, gap +0.002). *Results.* Room 55–81%. DKKM 0.480 against 0.400: fair gap +0.080 ($t$ 4.6, ten of ten); the premium across ROE quintiles is −0.24, +0.58, +0.41, +0.34, −0.54% a month; the U, $+0.2(\tilde{z}^2-1)$, gives +0.059 ($t$ 6.3). *Story.* Firms of middling productivity bear the shock; the most productive have the margin to absorb it and the least productive have already shed the capacity that carries it.

In each of the three the exposure is a function of the characteristics that no linear combination of them spans, and the premium is non-monotone in whatever the characteristics reveal about the firm; where it is monotone in something revealed (GS21's size ladder, leverage fixed by type, a one-signed U in size; Section 6), the linear model holds it however curved the exposure is.

## 5 Larger factor models

The three sections above change one feature of a structural model. This one asks whether more priced factors help, in a reduced form whose structure is fully controlled: a panel of firms with five characteristics, a market factor, and $K$ extra priced factors whose total price of risk is held fixed as $K$ grows. The only thing that varies across the rows is what the extra factors load on: a linear function of the observed characteristics, a product of two partly hidden firm states (Section 3b's interaction), or the square of one firm state (Section 3c's non-monotone exposure). The structural version, a second priced state in each model, is in Section 6.

**The reduced form.** A simulated panel (`variants/reduced/rf_model.py`) of $N = 500$ firms over $T = 500$ months after a 100-month burn-in, five seeds per cell, run through the same oracle ceilings and the same estimators as the structural economies at the 360-month window; its true conditional moments are known in closed form. Notation: $i$ indexes firms, $t$ months, $j = 1, \ldots, 5$ the characteristics (size, book-to-market, asset growth, profitability, momentum), $k = 1, \ldots, K$ the extra priced factors.

- **Latent processes.** Each firm carries five independent AR(1) processes $z_{ijt} = \rho_j z_{ij,t-1} + \sqrt{1-\rho_j^2}\,\eta_{ijt}$, $\eta$ standard normal, $\rho = (0.99, 0.98, 0.90, 0.95, 0.70)$, so each has unit stationary variance.
- **Observed characteristics.** $X_{ijt} = z_{ijt}$ in the `kfh` and `kfo` ladders. In the `lfh` ladder the five processes are the firm's true states and each characteristic is its state plus observation noise, $X_{ijt} = z_{ijt} + \sigma_\nu \nu_{ijt}$, with $\nu_{ijt}$ an independent AR(1) with the same $\rho_j$ and unit variance. The ladder's one noise parameter is $\sigma_\nu$: 0 (the states are observed exactly, rows `n0`) or 1 (noise as large as the state itself, rows `n1`). $\tilde{X}_{it}$ is $X_{it}$ standardized across firms at $t$. The oracle's bases and the estimators see the ranks of $X_{it}$, as everywhere in the report.
- **Returns.** $r_{i,t+1} = B_{it}\lambda + B_{it}f_{t+1} + \varepsilon_{i,t+1}$, with $f_{t+1} \sim N(0, \sigma_f^2 I_{K+1})$, $\sigma_f = 0.04$, and $\varepsilon_{i,t+1} \sim N(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon = 0.10$, all independent. $B_{it}$ is the $1 \times (K+1)$ row of firm $i$'s loadings and $\lambda$ the $(K+1)$-vector of constant prices of risk. The true premium is $\mu_{it} = B_{it}\lambda$ and the true covariance $\Sigma_t = \sigma_f^2 B_t B_t' + \sigma_\varepsilon^2 I$.
- **The market factor (column 0).** $B_{i0t} = 1 + a'\tilde{X}_{it}$ with $a = (0, 0.3, 0, 0.3, 0.3)$ on (size, bm, agr, roe, mom), and $\lambda_0 = 0.002$, a Sharpe of $\lambda_0/\sigma_f = 0.05$. This loading is linear in the characteristics, so every method holds it.
- **The $K$ extra factors (columns $k = 1, \ldots, K$).** $\lambda_k = 0.024/\sqrt{K}$ for every $k$, so each has Sharpe $0.6/\sqrt{K}$ and the total price of risk is the same at every $K$. Every extra loading is centered across

firms each month ($\bar{\beta}_{kt}$ is the cross-sectional mean), so the equal-weighted market carries none of the extra premia. The ladders differ only in the loading:

- ***kfh*** (loadings on observed characteristics): $B_{ikt} = h(w_k'\tilde{X}_{it}) - \bar{\beta}_{kt}$, with $w_k$ a fixed unit vector in $\mathbb{R}^5$ drawn once and reused in every seed, and $h(u) = u$ in the main table (lin); the appendix adds $h(u) = \tanh(u)$ (tanh) and $h(u) = u^2$ (sq). Exposure is a function of what the characteristics show directly: GS21's structure.
- ***kfo*** (Section 6): `kfh` with the size coordinate of each $w_k$ set to zero before normalizing, so no extra factor loads on size.
- ***lfh*** (loadings on products of latent states): $B_{ikt} = z_{ia_kt}\, z_{ib_kt} - \bar{\beta}_{kt}$, where $(a_k, b_k)$ is the $k$-th pair in a fixed enumeration of the ten pairs of the five processes (so $K \leq 10$). The loading is a product of two true states that the characteristics reveal only up to the observation noise $\sigma_\nu$: KP14's and BGN's structure, a type times a within-firm state.
- ***lsq, lsqb, lsqr*** (loading on the square of one state): $B_{ikt} = z_{iqt}^2 - 1$ for one process $q$, size, book-to-market or profitability, observed exactly (n0) or with noise $\sigma_\nu = 1$ (n1): Section 3c's structure, an exposure lowest at the mean of one characteristic and rising in both tails.

The ladders therefore vary one thing at a time: $K$; whether the loading is a function of the observed characteristics, a product of partly hidden states, or the square of one state; and how much noise hides the states.

Table 7: Reduced form: priced factors loading linearly on observed characteristics, on products of two partly hidden firm states, and on the square of one firm state (size, book-to-market or profitability). Five seeds per cell, the total price of risk fixed across K (per-factor price $\propto 1/\sqrt{K}$); K = number of priced factors; the lfh rows load products of pairs of the five latent processes, observed with noise of sd 0 (n0) or 1 (n1).

| configuration | $SR_{\max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kfh_lin_1 – K=1 factor on observed characteristics, linear loadings | 0.598 | 8 | 0.598 | +0.000 | 0 | 0.01 | 0.590 | 0.584 | 0.581 | 0.595 | 0.041 | -0.0067 (t -1.7) | 0/5 |
| kfh_lin_4 – K=4 | 0.592 | 8 | 0.588 | +0.003 | 0 | 0.06 | 0.555 | 0.578 | 0.567 | 0.575 | 0.041 | -0.0237 (t -3.1) | 0/5 |
| kfh_lin_16 – K=16 | 0.385 | 13 | 0.383 | +0.001 | 0 | 0.03 | 0.352 | 0.357 | 0.353 | 0.365 | 0.041 | -0.0128 (t -1.2) | 2/5 |
| lfh_L5_K1_n0 – K=1 product of latent states, no observation noise | 0.598 | 8 | 0.208 | +0.389 | 187 | 0.56 | 0.570 | 0.156 | 0.539 | 0.175 | 0.041 | +0.0311 (t 4.0) | 5/5 |
| lfh_L5_K2_n0 – K=2 | 0.598 | 8 | 0.144 | +0.453 | 314 | 0.58 | 0.564 | 0.111 | 0.520 | 0.115 | 0.041 | +0.0443 (t 6.7) | 5/5 |
| lfh_L5_K3_n0 – K=3 | 0.598 | 8 | 0.106 | +0.490 | 464 | 0.59 | 0.566 | 0.059 | 0.507 | 0.084 | 0.041 | +0.0589 (t 9.0) | 5/5 |
| lfh_L5_K5_n0 – K=5 | 0.598 | 8 | 0.092 | +0.502 | 549 | 0.59 | 0.543 | 0.048 | 0.423 | 0.064 | 0.041 | +0.1193 (t 21.6) | 5/5 |
| lfh_L5_K10_n0 – K=10 | 0.598 | 8 | 0.089 | +0.503 | 568 | 0.58 | 0.507 | 0.047 | 0.250 | 0.067 | 0.041 | +0.2572 (t 39.9) | 5/5 |
| lfh_L5_K1_n1 – K=1 product of latent states, observation noise 1.0 | 0.598 | 8 | 0.167 | +0.416 | 249 | 0.56 | 0.526 | 0.096 | 0.433 | 0.137 | 0.041 | +0.0930 (t 3.8) | 5/5 |
| lfh_L5_K2_n1 – K=2 | 0.598 | 8 | 0.111 | +0.469 | 423 | 0.57 | 0.523 | 0.072 | 0.420 | 0.082 | 0.041 | +0.1025 (t 6.8) | 5/5 |
| lfh_L5_K3_n1 – K=3 | 0.598 | 8 | 0.091 | +0.486 | 537 | 0.57 | 0.528 | 0.057 | 0.391 | 0.073 | 0.041 | +0.1371 (t 10.4) | 5/5 |
| lfh_L5_K5_n1 – K=5 | 0.598 | 8 | 0.075 | +0.496 | 664 | 0.57 | 0.496 | 0.038 | 0.309 | 0.060 | 0.041 | +0.1866 (t 26.3) | 5/5 |
| lfh_L5_K10_n1 – K=10 | 0.598 | 8 | 0.079 | +0.479 | 607 | 0.55 | 0.456 | 0.052 | 0.171 | 0.077 | 0.041 | +0.2854 (t 24.8) | 5/5 |
| lsq_a0_n0 – K=1, square of size, mean zero, state observed exactly | 0.598 | 8 | 0.175 | +0.422 | 242 | 0.57 | 0.557 | 0.153 | 0.565 | 0.151 | 0.041 | -0.0082 (t -1.5) | 2/5 |
| lsq_a0_n1 – observation noise 1.0 | 0.598 | 8 | 0.144 | +0.436 | 302 | 0.56 | 0.508 | 0.119 | 0.540 | 0.130 | 0.041 | -0.0313 (t -2.0) | 1/5 |
| lsqb_a0_n0 – K=1, square of book-to-market, mean zero, observed exactly | 0.598 | 8 | 0.199 | +0.398 | 200 | 0.56 | 0.564 | 0.142 | 0.125 | 0.173 | 0.041 | +0.3662 (t 11.0) | 5/5 |
| lsqb_a0_n1 – noise 1.0 | 0.598 | 8 | 0.190 | +0.392 | 206 | 0.55 | 0.515 | 0.120 | 0.092 | 0.146 | 0.041 | +0.3403 (t 6.5) | 5/5 |

Factors that load linearly on observed characteristics leave no room at any $K$ and a negative gap. Factors that load on products of partly hidden states give room and a gap that grow with $K$ and with the observation noise that hides the states, to +0.29 at ten factors with noise. A product of two mean-zero states has no additive part for a linear model to hold, and each extra factor adds another product.

A square is a U. The loading $z^2 - 1$ is lowest for firms at the mean of the characteristic and rises in both tails (a hump after a change of sign), the reduced-form twin of Section 3c. Squaring book-to-market (`lsqb`) gives room +0.40 and a fair gap of +0.37 ($t$ 11) with the state observed exactly, and +0.34 with observation noise as large as the state. Squaring size (`lsq`) gives the same room and no fair gap: market value is $e^{\text{size}}$, so the value-weighted market inside the Fama–French set holds size$^2$ mechanically and Fama–French's Sharpe is 0.57 of a maximum of 0.60. Squaring profitability (`lsqr`) is running; its rows join the table when its five seeds finish. Shifting the mean of the squared state, $(z + m)^2 - 1$, adds a linear part a linear portfolio holds and sends room to near zero at $m$ of 1 or more (Section 6), as the shifted products do for the interaction.

**The structural version: one mean-zero type ladder spread over $K$ priced states** (`vym3s`, `vym6s`; rows in the table of Section 6). $y \in \mathbb{R}^K$, $K$ independent OU states each priced $2.5/\sqrt{K}$; 24 mean-zero types with loading vectors of magnitude 0.07 or 0.14 at cosines $\pm 1$ to $\pm 0.20$ to the price vector, level-compensated as `vyg25s`. The linear ceiling is unchanged (0.19) but the nonlinear ceiling falls from 0.52 to 0.42: a type whose loading is misaligned with the price vector carries the full $y$ risk of its magnitude and earns only the cosine of the premium. Room is 110% of the linear ceiling against 201%, and the fair gap +0.100 ($t$ 21) and +0.099 ($t$ 15), ten of ten seeds at either $K$, half of `vyg25s`'s. $K = 3$ and $K = 6$ agree to the third digit: in the structural models, unlike the reduced form, more priced states do not multiply interactions, because every state acts through the one growth-option elasticity.

## 6 Additional economies

Everything else that was run. The table has the main tables' columns; the bullets describe each economy against its parent (notation as in Section 1). Three groups of them answer side questions: the crash added to the interaction economies instead of to a single-type economy (`bvyj25`, `gvyp10j`, `vyj25`), the GS21 product ladder (`gvyp05` to `gvyp20`, `gvyp10x`), and the earlier single-characteristic and price-of-risk wirings that the harmonized cells replace.

**BGN**

- `g0235f`: `bgnbase` with $\sigma_z \to g_s \sigma_z$, $s_t \in \{\text{calm}, \text{stress}\}$ a Markov chain, $g = 0.2$ calm and 3.5 stress; monthly switch probabilities 1/12 (calm to stress) and 1/6 (stress to calm), so stress is one third of months with a mean spell of 6 months.

- `g0235df`: the same $g$ with switch probabilities 1/6 and 1/12: stress two thirds of months, mean spell 12 months.

- `bvych0`: `bvy1ch` with three types, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated ($\theta_f = (V_0/V_f)^{1.2}$). Exposure is $b_f$ times the firm's mix of assets in place and growth option.

- `bvy25`: `bgnbase` plus $y$; type $f$'s project cash flows are $\theta_f C_j e^{b_f y_t}$, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated. Exposure is $b_f$ times the firm's mix of assets in place and growth option.

- `bvych41`: `bvych0` with $y$ on a 41-node chain instead of 21.

- `bvy9t2`: `bvych0` with a second independent 21-node chain and nine types, a pair $(f_1, f_2)$ drawn independently from $\{0.01, 0.035, 0.07\}$ for each chain, cash flows $C_j\, e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each: the nested nine-type design of `vy9t2` in BGN.

- `bvyc0`: `bvy25` with $\theta_f = 1$, so the type is visible in value.

- `bvyj25`: `bvych0` (group 3b) plus a crash: each month, with probability 1% whatever $y_t$, $y_{t+1}$ goes to the lowest node; otherwise the Tauchen transition. The kernel weights the crash transition by $e^2$, so its risk-neutral probability is about 7.4% a month before renormalization.

- `bvysg`: `bgnbase` plus $y$; type $f$'s project cash flows are $C_j e^{b_f y_t}$, $b_f \in \{-0.07, 0, +0.07\}$, not level-compensated.

- `bvychs`: `bvych0` with the ladder mirrored: six types of equal share, $b_f \in \{-0.07, -0.035, -0.01, 0.01, 0.035, 0.07\}$, level-compensated (bv_comp 1.2, $\theta_f$ 1 to 1.60 with $b = -0.07$ the reference), 21-node chain, price 2.5.

Table 8: Additional economies. Everything else that was run, by model, each described in the appendix against its parent; the RF rows are the reduced-form ladders with bounded (tanh), curvature (square) and size-orthogonal curvature (kfo) loadings, the sign ladder (lsh: the K=1 latent product with the factors' means shifted; lrt: a one-signed ratio), and the curvature-in-a-latent ladder (lsq / lsqb: the square of one partly hidden latent, size or book-to-market, with its mean shifted).

| model | economy | $\mathrm{SR}_{\max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `g0235f` – regime price ×0.2 / ×3.5, spells 4× faster | 0.237 | 62% | 0.163 | +0.045 | 28 | 0.13 | 0.169 | 0.141 | 0.131 | 0.161 | 0.146 | +0.0079 (t 4.8) | +0.05 | 10/10 |
| BGN | `g0235df` – stress 2/3 of the time, 4× faster | 0.489 | 17% | 0.432 | +0.036 | 9 | 0.18 | 0.423 | 0.392 | 0.351 | 0.428 | 0.077 | -0.0051 (t -2.2) | -0.05 | 2/10 |
| BGN | `bvy25` – three types on y, level-compensated (1.2) | 0.601 | 41% | 0.398 | +0.062 | 16 | 0.23 | 0.399 | 0.344 | 0.367 | 0.384 | 0.248 | +0.0148 (t 3.0) | +0.09 | 9/10 |
| BGN | `bvych41` – bvych0 on a 41-node chain | 0.565 | 38% | 0.336 | +0.052 | 15 | 0.19 | 0.328 | 0.279 | 0.291 | 0.320 | 0.215 | +0.0086 (t 2.8) | +0.06 | 8/10 |
| BGN | `bvyc0` – the same without compensation | 0.475 | 41% | 0.395 | +0.066 | 18 | 0.22 | 0.386 | 0.321 | 0.347 | 0.376 | 0.195 | +0.0104 (t 3.1) | +0.07 | 9/10 |
| BGN | `bvyj25` – the chain with a priced crash (1%/month to $-y_{\max}$, priced $e^2$) | 0.536 | 38% | 0.315 | +0.054 | 17 | 0.19 | 0.307 | 0.261 | 0.269 | 0.301 | 0.202 | +0.0066 (t 2.8) | +0.05 | 9/10 |
| BGN | `bvysg` – signed exposure ladder ($-0.07$ / 0 / +0.07) | 0.504 | 28% | 0.361 | +0.120 | 35 | 0.31 | 0.318 | 0.305 | 0.273 | 0.324 | 0.141 | -0.0199 (t -2.0) | -0.08 | 2/10 |
| GS21 | `g28` – countercyclical $\gamma(x)$, clipped | 0.314 | 95% | 0.312 | +0.001 | 0 | 0.02 | 0.303 | 0.254 | 0.256 | 0.305 | 0.298 | -0.0024 (t -4.3) | -0.04 | 0/10 |
| GS21 | `bx7` – two-state regime on $\gamma$, five types on x | 0.327 | 94% | 0.317 | +0.007 | 2 | 0.07 | 0.309 | 0.282 | 0.271 | 0.311 | 0.306 | -0.0017 (t -2.8) | -0.03 | 0/10 |
| GS21 | `gx7` – five types on productivity x | 0.485 | 98% | 0.474 | +0.008 | 2 | 0.08 | 0.473 | 0.393 | 0.383 | 0.473 | 0.473 | +0.0003 (t 1.6) | +0.01 | 5/10 |
| GS21 | `gvy1` – one type, output $\times e^{0.05y}$ | 0.636 | 95% | 0.633 | +0.002 | 0 | 0.05 | 0.626 | 0.608 | 0.598 | 0.629 | 0.602 | -0.0030 (t -4.0) | -0.05 | 1/10 |
| GS21 | `gvy1j` – crash-only factor: $e^{0.05y}$, the same crash | 0.379 | 96% | 0.377 | +0.001 | 0 | 0.03 | 0.372 | 0.350 | 0.341 | 0.374 | 0.363 | -0.0022 (t -4.4) | -0.04 | 1/10 |
| GS21 | `gvy25` – production-level exposure to y, three types | 0.725 | 86% | 0.709 | +0.014 | 2 | 0.14 | 0.688 | 0.682 | 0.666 | 0.695 | 0.625 | -0.0070 (t -4.0) | -0.09 | 1/10 |
| GS21 | `gvys25` – signed fixed-cost exposure to y | 0.619 | 69% | 0.602 | +0.014 | 2 | 0.13 | 0.496 | 0.490 | 0.426 | 0.499 | 0.420 | -0.0032 (t -1.5) | -0.03 | 2/6 |
| GS21 | `gvyp05` – product $\tilde{z}\tilde{b}$ at 0.05 | 0.425 | 60% | 0.284 | +0.128 | 45 | 0.30 | 0.297 | 0.254 | 0.202 | 0.276 | 0.254 | +0.0203 (t 7.4) | +0.11 | 10/10 |
| GS21 | `gvyp075` – product at 0.075 | 0.561 | 28% | 0.261 | +0.287 | 111 | 0.48 | 0.433 | 0.212 | 0.187 | 0.352 | 0.149 | +0.0802 (t 9.4) | +0.25 | 10/10 |
| GS21 | `gvyp15` – product at 0.15 | 0.744 | -25% | 0.616 | +0.119 | 22 | 0.39 | 0.665 | 0.349 | 0.516 | 0.612 | 0.192 | +0.0528 (t 9.3) | +0.26 | 10/10 |
| GS21 | `gvyp20` – product at 0.20 | 0.778 | -51% | 0.729 | +0.039 | 6 | 0.23 | 0.710 | 0.452 | 0.643 | 0.715 | 0.407 | -0.0049 (t -1.1) | -0.04 | 4/10 |
| GS21 | `gvyp10x` – product at 0.10 plus a curvature loading on x | 0.589 | 17% | 0.365 | +0.215 | 63 | 0.45 | 0.491 | 0.196 | 0.262 | 0.448 | 0.054 | +0.0431 (t 7.6) | +0.20 | 10/10 |
| GS21 | `gvyp10j` – the product exposure with a priced crash | 0.537 | 21% | 0.328 | +0.197 | 62 | 0.41 | 0.442 | 0.242 | 0.246 | 0.415 | 0.072 | +0.0267 (t 3.8) | +0.14 | 10/10 |
| GS21 | `gvypk2` – two priced chains, two product types | 0.478 | 27% | 0.226 | +0.212 | 103 | 0.37 | 0.309 | 0.168 | 0.153 | 0.254 | 0.105 | +0.0559 (t 3.2) | +0.15 | 9/10 |
| GS21 | `gvyzl` – productivity exposure, three fixed-coupon leverage types (diluting) | 0.865 | 50% | 0.793 | +0.059 | 8 | 0.31 | 0.804 | 0.793 | 0.743 | 0.799 | 0.426 | +0.0049 (t 2.5) | +0.05 | 7/10 |
| GS21 | `gvyzlf` – productivity exposure, three leverage types, capital structure fixed for life | 0.880 | 50% | 0.808 | +0.066 | 8 | 0.33 | 0.820 | 0.807 | 0.758 | 0.815 | 0.432 | +0.0046 (t 2.0) | +0.05 | 7/10 |
| GS21 | `gvyzlw` – gvyzlf with coupons 0.05 / 0.45 / 0.85 | 0.782 | 52% | 0.729 | +0.048 | 7 | 0.27 | 0.722 | 0.730 | 0.676 | 0.738 | 0.404 | -0.0165 (t -5.5) | -0.15 | 0/10 |
| GS21 | `gvyzl5` – gvyzlf with five leverage types | 0.880 | 50% | 0.812 | +0.060 | 7 | 0.32 | 0.818 | 0.812 | 0.765 | 0.818 | 0.440 | -0.0016 (t -0.7) | -0.03 | 4/10 |
| GS21 | `gvyzlb` – gvyzlf with the exposure doubled | 1.066 | 45% | 1.003 | +0.056 | 7 | 0.31 | 0.996 | 0.968 | 0.944 | 0.985 | 0.490 | +0.0114 (t 1.5) | +0.06 | 6/10 |
| GS21 | `gvyzlr` – gvyzlf with riskless debt | 0.880 | 49% | 0.808 | +0.066 | 8 | 0.33 | 0.820 | 0.807 | 0.758 | 0.815 | 0.432 | +0.0045 (t 2.0) | +0.05 | 7/10 |
| GS21 | `gvyzlp` – gvyzlf with a persistent y ($\kappa_y$ 0.10) | 0.475 | 76% | 0.457 | +0.017 | 4 | 0.13 | 0.448 | 0.428 | 0.418 | 0.451 | 0.355 | -0.0035 (t -2.7) | -0.04 | 2/10 |
| GS21 | `gvyzlrp` – gvyzlf with riskless debt and a persistent y | 0.475 | 76% | 0.457 | +0.017 | 4 | 0.13 | 0.448 | 0.428 | 0.419 | 0.451 | 0.356 | -0.0035 (t -2.7) | -0.04 | 2/10 |
| GS21 | `gsbg1` – benchmark at $\gamma_x$ 0.1 | 0.070 | 96% | 0.070 | +0.000 | 0 | 0.00 | 0.061 | 0.029 | 0.030 | 0.061 | 0.054 | +0.0004 (t 0.9) | +0.00 | 6/10 |
| GS21 | `gvyz10g` – gvyz10 at $\gamma_x$ 0.1 | 0.770 | 25% | 0.707 | +0.051 | 7 | 0.27 | 0.703 | 0.722 | 0.647 | 0.724 | 0.195 | -0.0229 (t -6.5) | -0.18 | 0/10 |
| GS21 | `gvyzlg` – gvyzlf at $\gamma_x$ 0.1 | 0.789 | 27% | 0.718 | +0.064 | 9 | 0.31 | 0.725 | 0.724 | 0.657 | 0.729 | 0.217 | -0.0051 (t -1.7) | -0.06 | 3/10 |
| GS21 | `gvyp10c` – the product with a leverage-level term, exposure 0.10 $\tilde{b}(\tilde{z}+2)$ | 0.774 | -86% | 0.763 | +0.006 | 1 | 0.09 | 0.745 | 0.727 | 0.720 | 0.767 | 0.674 | -0.0221 (t -4.1) | -0.17 | 0/10 |
| GS21 | `gvytl` – three exposure types × three fixed leverage types (KP14's recipe) | 0.739 | 84% | 0.728 | +0.009 | 1 | 0.12 | 0.708 | 0.688 | 0.685 | 0.714 | 0.621 | -0.0060 (t -3.7) | -0.07 | 2/10 |
| GS21 | `gvytlg` – the same at $\gamma_x$ 0.1 | 0.517 | 72% | 0.498 | +0.017 | 3 | 0.13 | 0.473 | 0.448 | 0.443 | 0.482 | 0.374 | -0.0089 (t -4.7) | -0.09 | 0/10 |
| BGN | `bvy9t2` – two priced chains, nine types (independent ladders), total price 2.5 | 0.636 | 42% | 0.398 | +0.063 | 16 | 0.23 | 0.395 | 0.339 | 0.355 | 0.383 | 0.266 | +0.0124 (t 3.5) | +0.08 | 8/10 |
| BGN | `bvychs` – bvych0 with sign-switching types: six types, loadings $\pm0.01, \pm0.035, \pm0.07$ | 0.510 | 36% | 0.349 | +0.069 | 20 | 0.23 | 0.360 | 0.274 | 0.277 | 0.334 | 0.183 | +0.0255 (t 4.9) | +0.12 | 9/10 |
| KP14 | `vyg25s` – vyg25 with sign-switching types: six types, loadings $\pm0.02, \pm0.07, \pm0.14$ | 0.575 | 11% | 0.174 | +0.349 | 226 | 0.49 | 0.373 | 0.102 | 0.190 | 0.153 | 0.056 | +0.1810 (t 17.2) | +0.32 | 10/10 |
| KP14 | `vyg25u` – vyg25 without the level compensation: the type shows in value at $y = 0$ | 0.495 | 11% | 0.351 | +0.066 | 19 | 0.22 | 0.339 | 0.290 | 0.293 | 0.341 | 0.044 | -0.0018 (t -0.5) | -0.02 | 3/10 |
| GS21 | `gvy25s` – gvy25 with sign-switching types: six level-exposure types, $\pm0.015, \pm0.05, \pm0.1$ | 0.619 | 52% | 0.554 | +0.061 | 11 | 0.27 | 0.549 | 0.541 | 0.506 | 0.560 | 0.318 | -0.0118 (t -4.4) | -0.10 | 2/10 |
| KP14 | `kpsgn` – the price of x switches sign with y: $\gamma_x(2\,\mathrm{logistic}(-2y) - 1)$ (conditioning room) | 0.143 | 42% | 0.075 | +0.050 | 90 | 0.10 | 0.060 | 0.018 | 0.045 | 0.064 | -0.002 | -0.0041 (t -0.9) | -0.01 | 1/3 |
| KP14 | `vym6` – six priced states, 24 rank-6 types (vym3's construction at K = 6) | 0.421 | 15% | 0.300 | +0.056 | 19 | 0.19 | 0.287 | 0.260 | 0.275 | 0.286 | 0.059 | +0.0002 (t 0.1) | -0.01 | 4/10 |
| KP14 | `vym3s` – vyg25s's signed ladder over three priced states: 24 mean-zero types, mirrored alignments | 0.503 | 13% | 0.191 | +0.224 | 126 | 0.37 | 0.296 | 0.134 | 0.195 | 0.171 | 0.065 | +0.0998 (t 20.9) | +0.22 | 10/10 |
| KP14 | `vym6s` – vyg25s's signed ladder over six priced states: 24 mean-zero types, mirrored alignments, orthogonal off-$\gamma$ directions | 0.507 | 13% | 0.192 | +0.225 | 124 | 0.37 | 0.296 | 0.132 | 0.194 | 0.168 | 0.065 | +0.0987 (t 15.0) | +0.22 | 10/10 |
| GS21 | `gvy2s` – gvy25s's signed ladder over two priced chains, cash-flow exposure only: twelve mean-zero types | 0.625 | 53% | 0.567 | +0.046 | 8 | 0.23 | 0.540 | 0.542 | 0.505 | 0.564 | 0.333 | -0.0246 (t -5.1) | -0.16 | 0/2 |
| GS21 | `gvytls` – gvytl with the exposure ladder mirrored: six signed exposure types × three fixed coupons, eighteen mean-zero types | 0.627 | 51% | 0.580 | +0.044 | 8 | 0.23 | 0.580 | 0.493 | 0.544 | 0.588 | 0.313 | -0.0074 (t -1.0) | -0.06 | 1/3 |
| BGN | `bvychs4` – bvychs at bv_comp 0.4 | 0.612 | 29% | 0.519 | +0.059 | 12 | 0.25 | 0.497 | 0.468 | 0.465 | 0.497 | 0.177 | -0.0035 (t -0.3) | -0.01 | 2/5 |
| BGN | `bvychs6` – bvychs at bv_comp 0.6 | 0.656 | 28% | 0.578 | +0.047 | 8 | 0.23 | 0.547 | 0.493 | 0.512 | 0.545 | 0.183 | +0.0013 (t 0.3) | +0.00 | 1/3 |
| BGN | `bvychs8` – bvychs at bv_comp 0.8 | 0.684 | 28% | 0.590 | +0.044 | 7 | 0.23 | 0.577 | 0.501 | 0.468 | 0.568 | 0.189 | +0.0086 (t 1.0) | +0.06 | 2/3 |
| KP14 | `vyg25s6` – vyg25s at bv_comp 0.6 | 0.576 | 11% | 0.212 | +0.305 | 147 | 0.47 | 0.337 | 0.183 | 0.131 | 0.185 | 0.062 | +0.1424 (t 24.2) | +0.27 | 5/5 |
| KP14 | `vyg25s18` – vyg25s at bv_comp 1.8 | 0.574 | 11% | 0.209 | +0.312 | 167 | 0.47 | 0.387 | 0.106 | 0.172 | 0.193 | 0.062 | +0.1854 (t 8.7) | +0.33 | 5/5 |
| GS21 | `gvyib2` – gvyib at gs_bib $-0.2$ (the $-1.0$ and $-0.5$ solves diverged: the firm picks the coupon that makes investment near-free) | 0.327 | 96% | 0.326 | +0.001 | 0 | 0.03 | 0.319 | 0.298 | 0.293 | 0.322 | 0.314 | -0.0027 (t -11.0) | -0.04 | 0/10 |
| GS21 | `gxq35` – gsbase with the x loading hump-shaped in productivity: $1 - 0.35\,(\tilde{z}^2 - 1)/\sqrt{2}$ | 0.284 | 96% | 0.282 | +0.002 | 1 | 0.03 | 0.270 | 0.248 | 0.242 | 0.270 | 0.269 | +0.0002 (t 0.4) | +0.00 | 1/3 |
| GS21 | `gxq70` – gsbase with the x loading hump-shaped in productivity: $1 - 0.70\,(\tilde{z}^2 - 1)/\sqrt{2}$ | 0.278 | 95% | 0.269 | +0.008 | 3 | 0.07 | 0.261 | 0.211 | 0.201 | 0.261 | 0.259 | +0.0006 (t 1.8) | +0.01 | 2/3 |
| GS21 | `gvy25sq` – gvy25s at $\gamma_v$ 0.25, bv_comp 12.0: the level shifts kept, the y price cut | 0.307 | 87% | 0.305 | +0.002 | 1 | 0.03 | 0.290 | 0.262 | 0.239 | 0.294 | 0.263 | -0.0038 (t -8.2) | -0.05 | 0/3 |
| GS21 | `gvy25sh` – gvy25s at $\gamma_v$ 0.5, bv_comp 9.0: the level shifts kept, the y price cut | 0.351 | 71% | 0.341 | +0.009 | 3 | 0.08 | 0.328 | 0.309 | 0.286 | 0.336 | 0.246 | -0.0077 (t -2.5) | -0.07 | 0/3 |
| BGN | `bvychs18` – bvychs at bv_comp 1.8 | 0.340 | 49% | 0.207 | +0.055 | 27 | 0.16 | 0.218 | 0.198 | 0.178 | 0.208 | 0.167 | +0.0064 (t 1.4) | +0.03 | 6/8 |
| BGN | `bvychs25` – bvychs at bv_comp 2.5 | 0.449 | 36% | 0.250 | +0.163 | 67 | 0.33 | 0.341 | 0.286 | 0.223 | 0.246 | 0.159 | +0.0482 (t 3.7) | +0.15 | 9/10 |
| BGN | `bvychs30` – bvychs at bv_comp 3.0 | 0.488 | 32% | 0.277 | +0.182 | 67 | 0.36 | 0.384 | 0.318 | 0.268 | 0.279 | 0.157 | +0.0626 (t 5.6) | +0.20 | 10/10 |
| BGN | `bvychs35` – bvychs at bv_comp 3.5 | 0.508 | 31% | 0.294 | +0.189 | 67 | 0.38 | 0.403 | 0.315 | 0.309 | 0.303 | 0.156 | +0.0712 (t 5.8) | +0.22 | 10/10 |
| BGN | `bvychs40` – bvychs at bv_comp 4.0 | 0.518 | 30% | 0.305 | +0.189 | 65 | 0.39 | 0.413 | 0.305 | 0.334 | 0.319 | 0.155 | +0.0590 (t 4.6) | +0.20 | 10/10 |
| BGN | `bvychs45` – bvychs at bv_comp 4.5 | 0.525 | 29% | 0.281 | +0.224 | 81 | 0.42 | 0.422 | 0.316 | 0.345 | 0.315 | 0.154 | +0.0638 (t 2.5) | +0.21 | 4/4 |
| BGN | `bvychsn5` – bvychs at bv_comp -0.5 | 0.367 | 35% | 0.254 | +0.080 | 35 | 0.21 | 0.229 | 0.202 | 0.207 | 0.228 | 0.127 | -0.0054 (t -0.9) | -0.01 | 6/10 |
| BGN | `bvyvee` – V-shaped loading across a visible level ladder: level 0.6..1.4, $b$ = +0.07, +0.00, -0.07, +0.00, +0.07 | 0.485 | 34% | 0.352 | +0.076 | 23 | 0.24 | 0.352 | 0.305 | 0.320 | 0.344 | 0.164 | +0.0071 (t 1.3) | +0.05 | 6/8 |
| GS21 | `gvyql10` – U-shaped exposure in the firm's own leverage: $+0.1\,(\tilde{b}^2 - 1)$, one type | 0.287 | -81% | 0.259 | +0.015 | 8 | 0.08 | 0.247 | 0.230 | 0.231 | 0.249 | 0.238 | -0.0021 (t -1.9) | -0.03 | 0/3 |
| GS21 | `gvyql20` – U-shaped exposure in the firm's own leverage: $+0.2\,(\tilde{b}^2 - 1)$, one type | 0.602 | -95% | 0.593 | +0.007 | 1 | 0.09 | 0.581 | 0.570 | 0.565 | 0.583 | 0.571 | -0.0020 (t -1.9) | -0.05 | 0/3 |
| GS21 | `gvyqh10` – hump-shaped exposure in the firm's own leverage: $-0.1\,(\tilde{b}^2 - 1)$, one type | 0.469 | -66% | 0.409 | +0.032 | 11 | 0.16 | 0.366 | 0.358 | 0.335 | 0.391 | 0.339 | -0.0259 (t -6.9) | -0.14 | 0/3 |
| KP14 | `vyg25sx2` – vyg25s with the loadings doubled: $\pm0.04, \pm0.14, \pm0.28$ | 0.640 | 6% | 0.158 | +0.437 | 343 | 0.57 | 0.456 | 0.206 | 0.105 | 0.134 | 0.027 | +0.2452 (t 11.1) | +0.40 | 10/10 |
| KP14 | `vyg25sx3` – vyg25s with the loadings tripled: $\pm0.06, \pm0.21, \pm0.42$ | 0.661 | 5% | 0.248 | +0.364 | 175 | 0.55 | 0.461 | 0.223 | 0.098 | 0.237 | 0.021 | +0.2177 (t 9.7) | +0.39 | 10/10 |
| GS21 | `gvyzz10` – U-shaped exposure in the firm's own productivity: $+0.1\,(\tilde{z}^2 - 1)$, one type | 0.621 | 43% | 0.356 | +0.224 | 60 | 0.45 | 0.498 | 0.455 | 0.312 | 0.489 | 0.252 | +0.0086 (t 0.7) | +0.02 | 3/5 |
| GS21 | `gvyzz20` – U-shaped exposure in the firm's own productivity: $+0.2\,(\tilde{z}^2 - 1)$, one type | 0.686 | 26% | 0.441 | +0.168 | 43 | 0.42 | 0.521 | 0.350 | 0.321 | 0.462 | 0.179 | +0.0589 (t 6.3) | +0.23 | 10/10 |
| GS21 | `gvyzzh10` – hump-shaped exposure in the firm's own productivity: $-0.1\,(\tilde{z}^2 - 1)$, one type | 0.639 | 25% | 0.364 | +0.250 | 70 | 0.49 | 0.480 | 0.325 | 0.255 | 0.400 | 0.152 | +0.0795 (t 4.6) | +0.25 | 10/10 |
| GS21 | `gvy25sc25` – gvy25s at bv_comp 2.5: strong level compensation | 0.712 | 33% | 0.542 | +0.168 | 31 | 0.46 | 0.635 | 0.585 | 0.630 | 0.655 | 0.231 | -0.0198 (t -2.8) | -0.14 | 1/6 |
| BGN | `bvyhx2` – hump-shaped loading across a mild level ladder 0.85..1.15: $b$ = -0.14, +0.00, +0.14, +0.00, -0.14 | 0.597 | 27% | 0.526 | +0.057 | 11 | 0.25 | 0.523 | 0.392 | 0.315 | 0.508 | 0.158 | +0.0150 (t 0.7) | +0.06 | 2/3 |
| BGN | `bvyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.551 | 28% | 0.349 | +0.155 | 44 | 0.36 | 0.375 | 0.295 | 0.280 | 0.340 | 0.153 | +0.0345 (t 3.9) | +0.14 | 10/10 |
| BGN | `bvyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.28, +0.00, +0.28, +0.00, -0.28 | 0.642 | 25% | 0.357 | +0.260 | 73 | 0.50 | 0.476 | 0.341 | 0.293 | 0.344 | 0.159 | +0.1186 (t 10.9) | +0.31 | 10/10 |
| KP14 | `vyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.632 | 7% | 0.207 | +0.377 | 197 | 0.54 | 0.431 | 0.217 | 0.136 | 0.178 | 0.031 | +0.2123 (t 12.5) | +0.37 | 10/10 |
| KP14 | `vyhx4` – hump-shaped loading across a mild level ladder 0.8..1.2: $b$ = -0.28, +0.00, +0.28, +0.00, -0.28 | 0.659 | 5% | 0.298 | +0.318 | 123 | 0.53 | 0.480 | 0.309 | 0.199 | 0.278 | 0.022 | +0.1664 (t 10.6) | +0.36 | 10/10 |
| GS21 | `gvyhx3` – hump-shaped loading across a mild level ladder 0.8..1.2 (gs_level): $b$ = -0.21, +0.00, +0.21, +0.00, -0.21 | 0.823 | 26% | 0.706 | +0.112 | 17 | 0.41 | 0.766 | 0.737 | 0.725 | 0.764 | 0.210 | +0.0023 (t 0.9) | +0.04 | 7/10 |
| BGN | `bvyq2` – two chains, the second loading quadratic in the first: $b_2 = -14\,(b_1^2 - \bar{b}_1^2)$, $b_1 = \pm0.01, \pm0.035, \pm0.07$ | 0.460 | 37% | 0.303 | +0.063 | 21 | 0.20 | 0.307 | 0.240 | 0.260 | 0.285 | 0.172 | +0.0223 (t 18.1) | +0.11 | 3/3 |
| BGN | `bvyq2x` – two chains, the second loading quadratic in the first: $b_2 = -7\,(b_1^2 - \bar{b}_1^2)$, $b_1 = \pm0.02, \pm0.07, \pm0.14$ | 0.523 | 30% | 0.375 | +0.070 | 19 | 0.24 | 0.359 | 0.321 | 0.308 | 0.366 | 0.158 | -0.0075 (t -1.2) | -0.05 | 1/3 |
| BGN | `bvyq2k` – two chains, the second loading quadratic in the first: $b_2 = -28\,(b_1^2 - \bar{b}_1^2)$, $b_1 = \pm0.01, \pm0.035, \pm0.07$ | 0.470 | 34% | 0.315 | +0.071 | 22 | 0.22 | 0.304 | 0.265 | 0.261 | 0.302 | 0.159 | +0.0026 (t 0.9) | +0.01 | 6/10 |
| KP14 | `vyg25su` – vyg25s without level compensation (bv_comp 0): $\pm0.02, \pm0.07, \pm0.14$ | 0.575 | 11% | 0.290 | +0.230 | 84 | 0.43 | 0.395 | 0.270 | 0.066 | 0.274 | 0.056 | +0.1162 (t 11.1) | +0.28 | 10/10 |
| KP14 | `vyg25sux2` – vyg25sx2 without level compensation (bv_comp 0): $\pm0.04, \pm0.14, \pm0.28$ | 0.640 | 6% | 0.185 | +0.411 | 500 | 0.55 | 0.482 | 0.284 | 0.088 | 0.229 | 0.027 | +0.1914 (t 6.5) | +0.37 | 10/10 |
| KP14 | `vyg25x2` – vyg25 (positive ladder) with the loadings x2: 0.04, 0.14, 0.28 | 0.468 | 7% | 0.335 | +0.074 | 22 | 0.23 | – | – | – | – | – | – | – | – |
| KP14 | `vyg25x3` – vyg25 (positive ladder) with the loadings x3: 0.06, 0.21, 0.42 | 0.475 | 6% | 0.358 | +0.063 | 17 | 0.22 | – | – | – | – | – | – | – | – |
| KP14 | `vyg35x2` – vyg25x2 (positive ladder x2) with the $y$ price raised to 3.5 | 0.531 | 5% | 0.373 | +0.079 | 21 | 0.26 | – | – | – | – | – | – | – | – |
| KP14 | `vyg25x2c` – vyg25x2 (positive ladder x2) with the level compensation strengthened to 2.5 | 0.489 | 7% | 0.359 | +0.074 | 20 | 0.24 | – | – | – | – | – | – | – | – |
| KP14 | `vyge` – the $y$ exposure on the profitability deviation, $b_f\,(\varepsilon - 1)$: three types 0.1 / 0.25 / 0.45 | 0.477 | 19% | 0.267 | +0.038 | 14 | 0.15 | 0.250 | 0.228 | 0.254 | 0.251 | 0.088 | -0.0048 (t -1.9) | -0.05 | 0/3 |
| KP14 | `vygel` – vyge with a visible level ladder, type_level 0.6 / 1.0 / 1.6 | 0.491 | 17% | 0.269 | +0.046 | 17 | 0.16 | 0.266 | 0.254 | 0.236 | 0.256 | 0.077 | +0.0049 (t 1.1) | +0.03 | 2/3 |
| KP14 | `vygesm` – exposure on the profitability deviation, mirrored types $\pm0.05$ / 0.1 / 0.15 with a level ladder 0.5..2.0: a product of two mean-zero factors | 0.369 | 20% | 0.232 | +0.022 | 9 | 0.10 | 0.209 | 0.203 | 0.189 | 0.217 | 0.074 | -0.0083 (t -1.4) | -0.04 | 1/3 |
| KP14 | `vygelb` – bounded exposure (cap 2) on the profitability deviation, mirrored types $\pm0.1$ / 0.25 / 0.45 with a level ladder 0.5..2.0 | 0.482 | 16% | 0.228 | +0.058 | 26 | 0.17 | 0.219 | 0.199 | 0.203 | 0.215 | 0.074 | +0.0024 (t 1.4) | +0.02 | 7/10 |
| KP14 | `vygelb7` – bounded exposure (cap 2), mirrored types $\pm0.2$ / 0.45 / 0.7 with a level ladder 0.5..2.0 | 0.535 | 14% | 0.231 | +0.085 | 37 | 0.22 | 0.225 | 0.202 | 0.208 | 0.218 | 0.073 | +0.0049 (t 2.1) | +0.03 | 8/10 |
| KP14 | `vygelb7g` – bounded exposure (cap 2), mirrored types $\pm0.2$ / 0.45 / 0.7, level ladder 0.5..2.0, $\gamma_v$ 4.0 | 0.677 | 11% | 0.235 | +0.120 | 51 | 0.27 | 0.238 | 0.208 | 0.208 | 0.222 | 0.073 | +0.0135 (t 4.2) | +0.07 | 10/10 |
| KP14 | `vygelbw` – bounded exposure (cap 2), mirrored types $\pm0.3$ / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_v$ 4.0 | 0.710 | 11% | 0.243 | +0.138 | 56 | 0.29 | 0.254 | 0.216 | 0.216 | 0.228 | 0.073 | +0.0215 (t 4.3) | +0.10 | 10/10 |
| KP14 | `vygelbx` – bounded exposure (cap 2), mirrored types $\pm0.5$ / 1.2 / 2.0, level ladder 0.5..2.0, $\gamma_v$ 4.0 | 0.727 | 11% | 0.254 | +0.150 | 58 | 0.31 | 0.273 | 0.227 | 0.216 | 0.241 | 0.073 | +0.0297 (t 4.2) | +0.12 | 10/10 |
| KP14 | `vygelbc` – bounded exposure (cap 2.5), mirrored types $\pm0.3$ / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_v$ 4.0 | 0.752 | 9% | 0.244 | +0.170 | 69 | 0.33 | 0.272 | 0.220 | 0.219 | 0.231 | 0.068 | +0.0369 (t 6.6) | +0.14 | 10/10 |
| KP14 | `vygelbc6` – bounded exposure (cap 2.5), mirrored types $\pm0.3$ / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_v$ 6.0 | 0.892 | 8% | 0.249 | +0.201 | 80 | 0.37 | 0.290 | 0.230 | 0.222 | 0.237 | 0.069 | +0.0487 (t 5.8) | +0.16 | 10/10 |
| KP14 | `vygelbcs` – bounded exposure (cap 2.5), mirrored types $\pm0.3$ / 0.7 / 1.2, level ladder 0.5..2.0, $\gamma_v$ 4.0, $\sigma_\epsilon$ 0.3 | 0.858 | 7% | 0.297 | +0.256 | 90 | 0.46 | 0.388 | 0.251 | 0.288 | 0.282 | 0.052 | +0.0913 (t 12.3) | +0.25 | 10/10 |
| GS21 | `gvylu30c`$^\ddagger$ – exposure U-shaped in leverage: fixed coupons 0.15 / 0.40 / 0.70, $b$ = 0.3, 0, 0.3, value-compensated | 1.219 | 99% | 1.213 | +0.005 | 0 | 0.11 | – | – | – | – | – | – | – | – |
| GS21 | `gvylu30`$^\ddagger$ – exposure U-shaped in leverage: fixed coupons 0.15 / 0.40 / 0.70, $b$ = 0.3, 0, 0.3, no compensation | 0.967 | 95% | 0.951 | +0.016 | 2 | 0.17 | – | – | – | – | – | – | – | – |
| GS21 | `gvyu4`$^\ddagger$ – U-shaped loading across a mild level ladder 0.8..1.2 (gs_level): $b$ = +0.28, 0, 0, 0, +0.28 | 0.917 | 85% | 0.893 | +0.020 | 2 | 0.18 | – | – | – | – | – | – | – | – |
| KP14 | `vyu5r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.35 at the ends, unexposed reference type | 0.342 | 14% | 0.267 | +0.050 | 19 | 0.17 | 0.254 | 0.219 | 0.205 | 0.253 | 0.036 | +0.0003 (t 0.2) | -0.00 | 4/10 |
| KP14 | `vyu4r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28 at the ends, unexposed reference type | 0.367 | 14% | 0.262 | +0.069 | 27 | 0.20 | 0.251 | 0.223 | 0.208 | 0.248 | 0.038 | +0.0025 (t 0.7) | +0.01 | 5/10 |
| KP14 | `vyu4` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28, 0, 0, 0, +0.28 | 0.371 | 13% | 0.263 | +0.072 | 28 | 0.20 | 0.256 | 0.216 | 0.209 | 0.246 | 0.040 | +0.0100 (t 1.5) | +0.04 | 7/10 |
| BGN | `bvyu6r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.42 at the ends, unexposed reference type | 0.768 | 49% | 0.592 | +0.137 | 24 | 0.42 | 0.616 | 0.479 | 0.533 | 0.582 | 0.373 | +0.0246 (t 2.6) | +0.13 | 8/10 |
| BGN | `bvyu5r` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.35 at the ends, unexposed reference type | 0.774 | 46% | 0.641 | +0.108 | 17 | 0.38 | 0.664 | 0.562 | 0.558 | 0.623 | 0.358 | +0.0295 (t 3.1) | +0.16 | 9/10 |
| BGN | `bvyu4` – U-shaped loading across a mild level ladder 0.8..1.2: $b$ = +0.28, 0, 0, 0, +0.28 | 0.382 | 31% | 0.246 | +0.111 | 54 | 0.25 | 0.241 | 0.157 | 0.181 | 0.227 | 0.118 | +0.0142 (t 1.3) | +0.06 | 7/10 |
| KP14 | `vyg25` – three types on y, price 2.5 (control) | 0.495 | 11% | 0.346 | +0.082 | 24 | 0.25 | 0.350 | 0.292 | 0.320 | 0.335 | 0.044 | +0.0148 (t 5.2) | +0.09 | 10/10 |
| GS21 | `gvyp10` – product exposure $\tilde{z}\tilde{b}$ at 0.10 (control) | 0.650 | 8% | 0.447 | +0.194 | 47 | 0.46 | 0.553 | 0.238 | 0.333 | 0.548 | 0.018 | +0.0057 (t 0.7) | +0.01 | 5/10 |
| BGN | `bvych0` – y as a 21-node chain, three types (control) | 0.576 | 37% | 0.341 | +0.055 | 16 | 0.20 | 0.334 | 0.285 | 0.292 | 0.326 | 0.216 | +0.0076 (t 2.7) | +0.05 | 8/10 |
| GS21 | `gvysl15`$^\ddagger$ – industry sign × leverage: two industries (gs_level 0.8 / 1.2) × fixed coupons 0.15 / 0.40 / 0.70, $b$ = sign × 0.15 × coupon/0.4 | 1.190 | 66% | 1.183 | +0.006 | 1 | 0.12 | – | – | – | – | – | – | – | – |
| GS21 | `gvysz2` – type sign × productivity $\tilde{z}$: mirrored types $\pm0.15$ / 0.3 / 0.5, gs_level ladder 0.5..2.0, no compensation | 0.847 | 17% | 0.603 | +0.208 | 37 | 0.54 | 0.678 | 0.536 | 0.488 | 0.619 | 0.151 | +0.0590 (t 11.5) | +0.27 | 10/10 |
| GS21 | `gvysz1` – type sign × productivity $\tilde{z}$: mirrored types $\pm0.1$ / 0.2 / 0.35, gs_level ladder 0.5..2.0, no compensation | 0.844 | 25% | 0.530 | +0.278 | 60 | 0.60 | 0.658 | 0.560 | 0.431 | 0.605 | 0.210 | +0.0533 (t 6.1) | +0.24 | 10/10 |
| BGN | `bvychsx2` – bvychs with the loadings x 2: $\pm0.02, \pm0.07, \pm0.14$ | 0.496 | 32% | 0.344 | +0.089 | 26 | 0.26 | 0.347 | 0.308 | 0.279 | 0.333 | 0.161 | +0.0072 (t 1.1) | +0.04 | 7/10 |
| BGN | `bvychsx3` – bvychs with the loadings x 3: $\pm0.03, \pm0.105, \pm0.21$ | 0.533 | 31% | 0.384 | +0.081 | 21 | 0.26 | 0.377 | 0.353 | 0.321 | 0.370 | 0.164 | +0.0024 (t 0.5) | +0.01 | 6/10 |
| GS21 | `gvy25sx2` – gvy25s with the loadings x 2: $\pm0.03, \pm0.1, \pm0.2$ | 0.817 | 41% | 0.721 | +0.093 | 13 | 0.38 | 0.814 | 0.777 | 0.796 | 0.814 | 0.344 | +0.0005 (t 0.1) | +0.01 | 2/3 |
| KP14 | `kpgam` – $\gamma_x \times$ logistic(y), 1 to 3 | 0.390 | 65% | 0.373 | +0.005 | 1 | 0.06 | 0.358 | 0.348 | 0.340 | 0.363 | 0.254 | -0.0053 (t -4.0) | -0.05 | 1/10 |
| KP14 | `vyx` – three types on y, price 1.8 | 0.421 | 14% | 0.321 | +0.055 | 17 | 0.19 | 0.313 | 0.284 | 0.302 | 0.310 | 0.054 | +0.0009 (t 0.3) | +0.02 | 7/10 |
| KP14 | `vyj25` – the same with a priced crash in y | 0.554 | 10% | 0.323 | +0.104 | 33 | 0.28 | 0.315 | 0.270 | 0.292 | 0.307 | 0.045 | +0.0080 (t 2.2) | +0.06 | 8/10 |
| KP14 | `vym2t3` – two priced states, three loading vectors | 0.475 | 12% | 0.326 | +0.090 | 28 | 0.26 | 0.317 | 0.280 | 0.301 | 0.308 | 0.049 | +0.0075 (t 2.1) | +0.05 | 8/10 |
| KP14 | `vy9t2f` – two priced states, nine types, each state priced 2.5 | 0.582 | 6% | 0.415 | +0.077 | 19 | 0.26 | 0.416 | 0.314 | 0.330 | 0.401 | 0.018 | +0.0149 (t 4.2) | +0.10 | 9/10 |
| KP14 | `vym3t3` – three priced states, three types | 0.443 | 14% | 0.285 | +0.087 | 30 | 0.24 | 0.279 | 0.246 | 0.266 | 0.273 | 0.061 | +0.0036 (t 0.8) | +0.02 | 6/10 |
| KP14 | `vym3` – three priced states, twenty rank-3 types | 0.418 | 15% | 0.298 | +0.056 | 19 | 0.19 | 0.286 | 0.259 | 0.278 | 0.287 | 0.059 | -0.0032 (t -1.7) | -0.02 | 5/10 |
| RF | `kfh_tanh_1` – K=1, bounded (tanh) loadings on observed characteristics | 0.592 | 8% | 0.592 | +0.001 | 0 | 0.03 | 0.579 | 0.578 | 0.570 | 0.588 | 0.041 | -0.0105 (t -1.9) | -0.10 | 0/5 |
| RF | `kfh_tanh_4` – K=4 | 0.584 | 9% | 0.574 | +0.007 | 1 | 0.09 | 0.541 | 0.564 | 0.544 | 0.560 | 0.041 | -0.0232 (t -2.9) | -0.16 | 0/5 |
| RF | `kfh_tanh_16` – K=16 | 0.469 | 11% | 0.378 | +0.044 | 12 | 0.19 | 0.346 | 0.357 | 0.345 | 0.359 | 0.041 | -0.0152 (t -1.4) | -0.10 | 1/5 |
| RF | `kfh_sq_1` – K=1, squares of a random index of the characteristics | 0.600 | 8% | 0.080 | +0.519 | 647 | 0.59 | 0.582 | 0.032 | 0.046 | 0.059 | 0.041 | +0.5161 (t 57.3) | +0.58 | 5/5 |
| RF | `kfh_sq_4` – K=4 | 0.601 | 8% | 0.097 | +0.501 | 515 | 0.59 | 0.573 | 0.042 | 0.458 | 0.074 | 0.041 | +0.1155 (t 22.6) | +0.35 | 5/5 |
| RF | `kfh_sq_16` – K=16 | 0.601 | 8% | 0.076 | +0.523 | 687 | 0.59 | 0.571 | 0.036 | 0.429 | 0.052 | 0.041 | +0.1429 (t 14.3) | +0.38 | 5/5 |
| RF | `kfo_sq_1` – K=1, squares, directions orthogonal to size | 0.600 | 8% | 0.080 | +0.519 | 648 | 0.59 | 0.582 | 0.032 | 0.048 | 0.059 | 0.041 | +0.5163 (t 57.7) | +0.58 | 5/5 |
| RF | `kfo_sq_4` – K=4 | 0.601 | 8% | 0.092 | +0.507 | 552 | 0.59 | 0.575 | 0.058 | 0.061 | 0.081 | 0.041 | +0.4833 (t 25.4) | +0.57 | 5/5 |
| RF | `kfo_sq_16` – K=16 | 0.602 | 8% | 0.079 | +0.520 | 656 | 0.59 | 0.575 | 0.015 | 0.031 | 0.053 | 0.041 | +0.5030 (t 21.3) | +0.57 | 5/5 |
| RF | `lsh_a0_b0_n0` – K=1 product of two mean-zero latent states (sign ladder origin), no noise | 0.598 | 8% | 0.204 | +0.393 | 192 | 0.56 | 0.569 | 0.154 | 0.538 | 0.175 | 0.041 | +0.0309 (t 3.6) | +0.18 | 5/5 |
| RF | `lsh_a0_b1_n0` – one factor's mean shifted 1 sd | 0.598 | 8% | 0.594 | +0.004 | 1 | 0.07 | 0.582 | 0.579 | 0.577 | 0.593 | 0.041 | -0.0110 (t -1.8) | -0.11 | 1/5 |
| RF | `lsh_a0_b2_n0` – one factor shifted 2 sd | 0.598 | 8% | 0.597 | +0.001 | 0 | 0.03 | 0.587 | 0.582 | 0.578 | 0.596 | 0.041 | -0.0094 (t -1.9) | -0.10 | 0/5 |
| RF | `lsh_a1_b1_n0` – both factors shifted 1 sd | 0.598 | 8% | 0.596 | +0.002 | 0 | 0.05 | 0.588 | 0.582 | 0.584 | 0.592 | 0.041 | -0.0048 (t -2.0) | -0.07 | 0/5 |
| RF | `lsh_a2_b2_n0` – both 2 sd | 0.598 | 8% | 0.597 | +0.001 | 0 | 0.02 | 0.590 | 0.583 | 0.583 | 0.594 | 0.041 | -0.0047 (t -1.9) | -0.07 | 0/5 |
| RF | `lsh_a3_b3_n0` – both 3 sd | 0.598 | 8% | 0.598 | +0.000 | 0 | 0.02 | 0.591 | 0.583 | 0.583 | 0.595 | 0.041 | -0.0046 (t -1.9) | -0.07 | 0/5 |
| RF | `lsh_a0_b0_n1` – mean-zero product, observation noise 1.0 | 0.598 | 8% | 0.160 | +0.422 | 263 | 0.56 | 0.524 | 0.094 | 0.431 | 0.133 | 0.041 | +0.0928 (t 3.7) | +0.30 | 5/5 |
| RF | `lsh_a2_b2_n1` – both shifted 2 sd, noise 1.0 | 0.598 | 8% | 0.593 | +0.001 | 0 | 0.03 | 0.583 | 0.582 | 0.572 | 0.589 | 0.041 | -0.0060 (t -2.1) | -0.08 | 0/5 |
| RF | `lrt_n0` – one-signed coverage ratio of the two latents, no noise | 0.598 | 8% | 0.597 | +0.001 | 0 | 0.04 | 0.591 | 0.579 | 0.571 | 0.593 | 0.041 | -0.0033 (t -2.3) | -0.05 | 0/5 |
| RF | `lsq_a1_n0` – square of size, the state shifted 1 sd | 0.598 | 8% | 0.596 | +0.002 | 0 | 0.05 | 0.585 | 0.578 | 0.581 | 0.595 | 0.041 | -0.0101 (t -1.9) | -0.11 | 0/5 |
| RF | `lsq_a2_n0` – shifted 2 sd | 0.598 | 8% | 0.598 | +0.000 | 0 | 0.02 | 0.588 | 0.580 | 0.579 | 0.597 | 0.041 | -0.0089 (t -1.9) | -0.10 | 0/5 |
| RF | `lsqb_a1_n0` – square of book-to-market, the state shifted 1 sd | 0.598 | 8% | 0.596 | +0.002 | 0 | 0.05 | 0.590 | 0.582 | 0.577 | 0.593 | 0.041 | -0.0027 (t -2.2) | -0.06 | 0/5 |
| RF | `lsqb_a1_n1` – shifted 1 sd, noise 1.0 | 0.598 | 8% | 0.589 | +0.002 | 0 | 0.05 | 0.581 | 0.579 | 0.560 | 0.585 | 0.041 | -0.0047 (t -1.9) | -0.07 | 1/5 |
| RF | `lsqb_a2_n0` – shifted 2 sd | 0.598 | 8% | 0.597 | +0.000 | 0 | 0.02 | 0.592 | 0.583 | 0.580 | 0.594 | 0.041 | -0.0026 (t -2.3) | -0.05 | 0/5 |
| RF | `lsqb_a2_n1` – shifted 2 sd, noise 1.0 | 0.598 | 8% | 0.593 | +0.001 | 0 | 0.03 | 0.584 | 0.582 | 0.568 | 0.589 | 0.041 | -0.0047 (t -1.9) | -0.07 | 0/5 |

- `bvychs18`, `bvychs25`, `bvychs30`, `bvychs35`, `bvychsn5`: `bvychs` at bv_comp 1.8 / 2.5 / 3.0 / 3.5 / −0.5. Ten seeds each (bvychs18 eight). bgn_vy's SDF step had its K4 pair sum rewritten as a blocked sum (commit 78961c4) after these ran out of memory at 36 GB; the arithmetic is unchanged to $3 \times 10^{-15}$.

- `bvyvee`: five types of equal share on a visible level ladder, $\theta_f = 0.6$ / 0.8 / 1.0 / 1.2 / 1.4 multiplying the cash-flow level, exposure V-shaped across it, $b_f = +0.07$, 0, −0.07, 0, +0.07; bv_comp 1.0, price 2.5. Eight seeds; two stopped. `bvyhump`, its mirror, was solved and not simulated.

- `bvychsx2`, `bvychsx3`: `bvychs` with the loadings ×2 (±0.02, ±0.07, ±0.14) and ×3 (±0.03, ±0.105, ±0.21). Ten seeds each.

- `bvychs40`, `bvychs45`: `bvychs` at bv_comp 4.0 / 4.5. Ten seeds; four seeds, stopped.

- `bvyhx2`, `bvyhx3`, `bvyhx4`: five types on a mild visible level ladder ($\theta_f$ 0.85–1.15 / 0.80–1.20 / 0.80–1.20) with the loading hump-shaped across it, $b_f = -c, 0, +c, 0, -c$, $c = 0.14$ / 0.21 / 0.28; bv_comp 1, price 2.5. Three seeds (stopped, room 9–13%); ten; ten.

- `bvyu4`, `bvyu5`, `bvyu6`, `bvyu5r`, `bvyu6r`: the hump ladder's chassis with a U-shaped loading, $+c$ at the two ends, $c = 0.28$ / 0.35 / 0.42; `bvyu5` failed the gate (the small exposed type is the compensation's reference, $J^* = 0$), `bvyu6` was not built, the `r` versions list the unexposed level-1.0 type first. Ten seeds each.

- `bvyq2`, `bvyq2x`, `bvyq2k`: `bvychs`'s six types with a second priced chain (both at $2.5/\sqrt{2}$) whose loading is quadratic in the first, $b_2 = \kappa\,(b_1^2 - \overline{b_1^2})$: $\kappa = -14$ on ±0.01/0.035/0.07; −7 on ±0.02/0.07/0.14; −28 on ±0.01/0.035/0.07. Three seeds each (stopped); ten.

**GS21**

- `g28`: `gsbase` with $\gamma_x \to \gamma(x_t) = \text{clip}\big(0.5 - 0.28\,x_t/\text{sd}(x),\ 0.05,\ 1.0\big)$, countercyclical.

- `bx7`: `gsbase` with $\gamma_x \to g_s\gamma_x$, $g \in \{0.6, 3.0\}$ on a two-state Markov regime, and five types with output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$.

- `gvyp10`: `gsbase` plus $y$ (9-node chain), one type, output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$, with $\tilde{z} = z/\text{sd}(z)$ and $\tilde{b} = (b - 0.5)/0.29$ the standardized coupon. Exposure is productivity × leverage, revealed by profitability and leverage.

- `gx7`: `gsbase` with five types, output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$, and $\gamma(x_t) = \text{clip}\big(0.5 - 0.4\,x_t/\text{sd}(x),\ 0.05,\ 1.5\big)$. A firm's exposure to the one priced shock is its type, which size and profitability reveal.

- `gvy1`: `gsbase` plus $y$ (9-node chain over ±2.5 sd), one type; output $e^{x_t + z_{it} + 0.05\,y_t} k_{it}$, a level exposure.

- `gvy1j`: `gvy1` with $\gamma_v = 0$ and the crash, on a chain over ±3.5 sd.

- `gvy25`: `gsbase` plus $y$ (9-node chain); output $e^{x_t + z_{it} + b_f y_t + a_f} k_{it}$, $b_f \in \{0.015, 0.05, 0.10\}$, with $a_f$ the level compensation.

- `gvy25s`: `gvy25` with the ladder mirrored: six types of equal share, $b_f \in \{-0.10, -0.05, -0.015, 0.015, 0.05, 0.10\}$, level-compensated (bv_comp 1.2) at price 2.5.

- `gvys25`: `gsbase` plus $y$, entering through an operating cost: $0.2\,e^{c_f y_t}$ per unit capital is subtracted from output, $c_f \in \{+0.4, 0, -0.4\}$; output itself does not load on $y$.

- `gvyp05, gvyp075, gvyp15, gvyp20`: `gvyp10` with the coefficient 0.10 replaced by 0.05, 0.075, 0.15 and 0.20.

- `gvyp10x`: `gvyp10` with output $e^{(1+(\tilde{z}_{it}^2-1)/\sqrt{2})\,x_t+z_{it}+0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t}k_{it}$: the loading on $x$ is also nonlinear in productivity.

- `gvyp10j`: `gvyp10` (group 3b) plus the same priced crash on its 9-node chain.

- `gvypk2`: `gvyp10` with a second independent chain (5 nodes each, 25 joint); two types of equal share with output $e^{x_t+z_{it}+0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t}k_{it}$ and $e^{x_t+z_{it}+0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t^{(2)}}k_{it}$.

- `gvyzl`: `gvyz10` with three leverage types: the coupon the firm refinances to is fixed at 0.158 / 0.421 / 0.684 per unit capital, equal shares drawn independently of productivity; refinancing opportunities (1% a month) and dilution at investment as in GS21, so the realized coupon drifts below the type's.

- `gvyzlf`: `gvyzl` with the capital structure fixed for life: no refinancing, new debt issued pari passu at investment (debt-to-equity 0.25 / 1.0 / 3.25).

- `gvyzlw`: `gvyzlf` with coupons 0.05 / 0.45 / 0.85; the high type is distressed (premium 16% a year, default rate 1.4%).

- `gvyzl5`: `gvyzlf` with five types, coupons 0.10 to 0.70.

- `gvyzlb`: `gvyzlf` with the exposure doubled to $0.20\,\tilde{z}\,y$.

- `gvyzlr`: `gvyzlf` with riskless debt: the debt claim is worth its riskless coupon value in every state (creditors made whole on default).

- `gvyzlp`: `gvyzlf` with $\kappa_y = 0.10$ a year instead of 0.35.

- `gvyzlrp`: both.

- `gsbg1`: the GS21 benchmark with $\gamma_x = 0.1$ instead of 0.5 (`gs_vy` at one $y$ node and no exposure, which is `gsbase`'s solver).

- `gvyz10g`: `gvyz10` at $\gamma_x = 0.1$.

- `gvyzlg`: `gvyzlf` at $\gamma_x = 0.1$.

- `gvyp10c`: `gvyp10` with a leverage-level term, output $e^{x+z+0.10\,\tilde{b}(\tilde{z}+2)\,y}$: a levered firm's exposure keeps leverage's sign for every firm with $\tilde{z} > -2$. The mean firm ($\tilde{b} < 0$) is a hedge against $y$, so the market's premium is negative.

- `gvytl`: KP14's 3b recipe in GS21: three latent exposure types, output $e^{x+z+b_f y+a_f}$ with $b_f \in \{0.015, 0.05, 0.10\}$ and $a_f$ the level compensation, times three fixed leverage types (coupons 0.158 / 0.421 / 0.684, no refinancing), nine types of equal share.

- `gvytlg`: `gvytl` at $\gamma_x = 0.1$ (benchmark `gsbg1`).

- `gxq35`, `gxq70`: `gsbase` (no $y$) with the productivity loading $1 + c\,(\tilde{z}^2 - 1)/\sqrt{2}$, $c = -0.35$ / $-0.70$. Ten oracle seeds, three estimator seeds; stopped.

- `gvy25sq`, `gvy25sh`: `gvy25s` at $\gamma_v$ 0.25 / 0.5 with bv_comp 12 / 9 (the types' level shifts kept, the $y$ price cut). Ten oracle seeds, three estimator seeds; stopped.

- `gvyql10`, `gvyql20`, `gvyqh10`: one type on `gvy25s`'s chassis with the $y$ exposure $c\,(\tilde{b}^2 - 1)$, $\tilde{b}$ the standardized coupon, $c = +0.1$ / $+0.2$ / $-0.1$. Three seeds each; stopped.

- `gvy25sx2`: `gvy25s` with the loadings $\times 2$ ($\pm 0.03$, $\pm 0.10$, $\pm 0.20$). Five oracle seeds, three estimator seeds; stopped. `gvy25sx3`: the $b_f = +0.045$ type's solve oscillated to the sweep cap; not run.

- `gvyzz10`, `gvyzz20`, `gvyzzh10`: one type on `gvy25s`'s chassis with the $y$ exposure $c\,(\tilde{z}^2 - 1)$, $\tilde{z}$ the standardized idiosyncratic productivity, $c = +0.1$ / $+0.2$ / $-0.1$ (`gs_bzz`, new). Five (stopped), ten and ten seeds.

- `gvyhx3`, `gvyhx4`, `gvyu4`: the size ladder in GS21 through the new gs_level (a per-type production-level multiplier; 1.0 leaves every earlier economy unchanged): five types at 0.8–1.2, bv_comp 1.0, NY 9, y_mult 2.5, price 2.5, one solve per type; the hump $\pm 0.21$ / $\pm 0.28$ and the U $+0.28$. gvyhx3 ten seeds (no gap), gvyhx4 solved and not pursued, gvyu4 stopped at the oracle.

- `gvylu30`, `gvylu30c`: the exposure U-shaped in leverage on `gvyzlf`'s fixed-coupon chassis (coupons 0.15 / 0.40 / 0.70 for life), production exposure gs_bv (0.3, 0, 0.3), without and with the $y = 0$ value compensation; both stopped at the oracle.

- `gvysz1`, `gvysz2`: the industry sign times the firm's standardized productivity (gs_bzl 0, gs_bzi 1), six mirrored types $\pm 0.1/0.2/0.35$ and $\pm 0.15/0.3/0.5$ on the gs_level ladder 0.5–2.0, bv_comp 0, price 2.5, six solves each; ten seeds each.

- `gvysl15`, `gvysl20`: the industry sign times financial leverage, two industries (gs_level 0.8 / 1.2) times three fixed coupons, exposure sign $\times c \times$ coupon/0.40 at $c = 0.15$ / $0.20$, bv_comp 1.0; gvysl15 stopped at the oracle, gvysl20 solved and not run.

- `gvy25sc25`: `gvy25s` at bv_comp 2.5. Six seeds; stopped. `gvy25sc32`: one type's solve cycle-capped; not run.

### KP14

- `kpgam`: `kpbase` with $\gamma_x \to \gamma_x\, g(y_t)$, $g(y) = 1 + 2/(1 + e^{2y})$, rising from 1 when $y$ is high to 3 when it is low; $y$ is unpriced ($\gamma_v = 0$) and no firm is exposed to it.

- `vyg25`: `vy1` with three types, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated. Growth options are options on the $e^{b_f y}$ cash flows, so a firm's exposure is $b_f$ times the elasticity of its value, which rises with its growth-option share: type × growth-option share, revealed by size and book-to-market.

- `vyx`: `kpbase` plus $y$ with $\gamma_v = 1.8$; type $f$'s project cash flows are $\theta_f e^{b_f y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated.

- `vyj25`: `vyg25` (group 3b) plus a Poisson jump of $y$ to $-y_{\max} = -3.5$ at intensity 0.12 a year (1% a month) under $P$ and $0.12\, e^2 = 0.89$ a year under $Q$.

- `vym3t3`: `vyg25` with $y \in \mathbb{R}^3$, three independent OU states; type $f$'s cash flows load $e^{b_f' y_t}$ with three vectors $b_f \in \mathbb{R}^3$ spanning the three states.

- `vym2t3`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; three types with loading vectors $(0.02, 0)$, $(0, 0.07)$, $(0.099, 0.099)$, price $2.5/\sqrt{2}$ on each: three sorts in three directions, which prices types 1 and 2 at an effective 1.77 instead of 2.5. The group-4 cell until October 21, 2026.

- `vy9t2f`: `vy9t2` with each state priced at 2.5 instead of $2.5/\sqrt{2}$; nests `vyg25` exactly.
- `vyg25s`: `vyg25` with the ladder mirrored: six types of equal share, $b_f \in \{-0.14, -0.07, -0.02, 0.02, 0.07, 0.14\}$, level-compensated as vyg25 (bv_comp 1.2, $\theta_f$ 1 to 3.36 with $b = -0.14$ the reference), price 2.5.
- `vyg25u`: `vyg25` with bv_comp 0: no level compensation, so the type shows in value at $y = 0$. The returns are vyg25's exactly ($\mathrm{SR}_{\max}$ equal to every digit); only the characteristics change.
- `vym3`: the same three states with twenty types, $b_f \in \mathbb{R}^3$ of rank 3.

What they add to the main results:

- **The chain's resolution is not what separates `bvych0` from `bvy25`**: on 41 nodes the gap is +0.009 against +0.008 on 21, and the difference from the Gaussian economy's +0.015 (sd 0.016) is within noise.
- **A level exposure of GS21 output to $y$** (`gvy1`, `gvy1j`) has no room at all, as every other level wiring in GS21; the productivity exposure (`gvyz10`) has 5% and the product with the coupon 43%.
- **The crash on an interaction economy** leaves room and gap at the control's in BGN (+0.007 against +0.008) and GS21 (+0.030 against +0.020, within noise) and halves KP14's (+0.008 against +0.014): the crash months are a state the ranks-times-state linear benchmark reads as well as DKKM does.
- **The GS21 product ladder**: room is hump-shaped in the product's size (45 / 111 / 47 / 22 / 6% of the linear ceiling at 0.05 / 0.075 / 0.10 / 0.15 / 0.20) while the gap is +0.019 / +0.077 / +0.020 / +0.078 / +0.022; above 0.10 the market becomes a hedge. A second nonlinear loading on $x$ (`gvyp10x`) gives +0.049: a second interaction factor adds. `bvy25`, the Gaussian-$y$ version of `bvych0`, has twice its gap at the same room, and the visibility ladder (`bvy25`, `bvyc0`, and `bvyc10` at `bv_comp` 1.0: +0.015 / +0.010 / +0.010) is flat within noise.
- **Two apparent failures of the mechanism are estimation cost, not structure.** `vyx` is `vyg25` at a lower price of $y$: its room (0.054) is about what DKKM gives up to estimation relative to the linear model (0.053), so the gap is zero, and the higher price lifts the room more than the cost. `bvysg` has twice `bvy25`'s room but DKKM's estimation loss is two and a half times larger (0.16 against 0.06): with signed exposures the nonlinear part of the tangency portfolio is a direct bet on the persistent state $y$, whose sample mean a 360-month window barely pins down. `bvychs` (appendix), the signed ladder with six types and the level compensation, has a positive gap of +0.026 at room 20%, so the loss in `bvysg` came with the visible type, as in `vyg25u`. What matters is room net of DKKM's estimation cost. Four more GS21 wirings (investment cost, floating coupon, a larger option, state-dependent recovery) were stopped at the oracle stage with room below +0.01, and GS21's default channel is dormant at every calibration tried.
- **Knobs that are not mechanisms**: the speed of the priced state ($\kappa$ 0.15 / 0.35 / 0.70 in KP14: +0.007 / +0.014 / +0.009, a mild hump within noise), the visibility of the type, the number of types (`vym3` against `vym3t3`), the sample size (`vym3` at $N = 1000$: room up, gap flat), and the ridge grid (DKKM +0.0001). The price of the interacted component is the one scaling lever (`vyx` to `vyg25`).
- `vyg25sx2`, `vyg25sx3`: `vyg25s` with the loadings ×2 (±0.04, ±0.14, ±0.28; $\theta_f$ 1 to 11.2) and ×3 (±0.06, ±0.21, ±0.42; $\theta_f$ 1 to 37). Ten seeds each.

- `vyg25su`, `vyg25sux2`: `vyg25s` and `vyg25sx2` at bv_comp 0 (the parents' returns; the type revealed by value). Ten seeds each.

- `vyge`, `vygel`, `vygels`, `vygesm`: the $y$ exposure on the profitability deviation (`bv_on_eps`, cash flow $xK^{\alpha}\theta u[1+(\varepsilon-1)e^{b_f y}]$): three types 0.1 / 0.25 / 0.45 with $\theta_f = 1$; the same with `type_level` 0.6 / 1.0 / 1.6; six mirrored types ±0.1/0.25/0.45 on a level ladder 0.5–2.0 (solved, not run: negative unit values for $b_f \leq -0.25$); ±0.05/0.1/0.15 on the same ladder. Three seeds each; stopped at the oracle.

- `vygelb`, `vygelb7`, `vygelb7g`, `vygelbw`, `vygelbx`, `vygelbc`, `vygelbc6`, `vygelbcs`: the bounded exposure on the profitability deviation (`bv_eps_cap` $c$: $h_b(y) = c/(1+(c-1)e^{-cby/(c-1)})$ in place of $e^{b_f y}$), six mirrored types on the level ladder 0.5–2.0, bv_comp 1.2: ±0.1/0.25/0.45 (cap 2, $\gamma_v$ 2.5); ±0.2/0.45/0.7 ($\gamma_v$ 2.5 and 4.0); ±0.3/0.7/1.2 ($\gamma_v$ 4.0; cap 2 and 2.5); ±0.5/1.2/2.0 ($\gamma_v$ 4.0); cap 2.5 at $\gamma_v$ 6.0 or $\sigma_\varepsilon$ 0.3. Ten seeds each.

- `vyhx3`, `vyhx4`, `vyu4`, `vyu4r`, `vyu5r`: BGN's size ladder in KP14 through type_level: five types at 0.8–1.2, bv_comp 1.0, price 2.5; the hump ±0.21 / ±0.28 and the U +0.28 (vyu4r / vyu5r with the unexposed level-1.0 type first, +0.28 / +0.35). Ten seeds each.

- `vyg25x2`, `vyg25x3`, `vyg35x2`, `vyg25x2c`: `vyg25` with the loadings ×2 / ×3, and ×2 at $\gamma_v$ 3.5 / bv_comp 2.5. Three to six oracle seeds; stopped.

### Two priced states

A second independent priced state, with the total price held at the one-state value ($2.5/\sqrt{2}$ on each). In BGN and GS21 three types span the two states with loading vectors in the pattern $(1,0)$, $(0,1)$, $(1,1)/\sqrt{2}$, at the 3b exposure sizes; in GS21 each type carries the 0.10 product exposure to its state. In KP14 the cell is the nested extension of 3b: every firm draws a type for each state independently from the 3b ladder $(0.02, 0.07, 0.14)$, nine types in all, so exposure to each state is its own type times leverage and zeroing the second ladder recovers `vyg25`. The three-direction KP14 version (`vym2t3`), the nested version with each state priced at the full 2.5 (`vy9t2f`) and the nested BGN version (`bvy9t2`) are in the appendix.

Table 9: Group 4. Larger factor models: two priced states. Two independent priced states at total price 2.5 ($2.5/\sqrt{2}$ each). BGN and GS21: three types with loading vectors (1,0), (0,1), (1,1)/$\sqrt{2}$. KP14: nine types, an independent draw from the 3b ladder for each state (the three-direction version and the full-price version are in the appendix).

| model | economy | $SR_{max}$ | market share | linear ceiling | room | room % of linear | interaction SR | DKKM | FMR | FF | linear | CAPM | fair gap | DKKM ⊥ | + |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BGN | `bvyk2` – two priced chains, three loading vectors | 0.639 | 36% | 0.393 | +0.071 | 18 | 0.25 | 0.394 | 0.332 | 0.353 | 0.379 | 0.228 | +0.0152 (t 4.3) | +0.10 | 10/10 |
| GS21 | `gvypk3` – two priced chains, three product types | 0.612 | 8% | 0.349 | +0.226 | 65 | 0.45 | 0.452 | 0.242 | 0.300 | 0.398 | 0.035 | +0.0515 (t 3.4) | +0.18 | 9/10 |
| KP14 | `vy9t2` – two priced states, nine types (independent ladders), total price 2.5 | 0.506 | 9% | 0.387 | +0.056 | 15 | 0.21 | 0.378 | 0.326 | 0.337 | 0.373 | 0.031 | +0.0048 (t 2.3) | +0.04 | 7/10 |

**What changes.** Total price fixed, $\gamma_v = 2.5/\sqrt{2}$ on each.

- `bvyk2`: `bvych0` with a second independent 21-node chain $y^{(2)}$; project cash flows

$$C_j\, e^{b_f y_t + b_f^{(2)} y_t^{(2)}}, \qquad (b_f, b_f^{(2)}) \in \{(0.01, 0),\ (0, 0.035),\ (0.049, 0.049)\}.$$

- `gvypk3`: `gvyp10` with a second independent chain (5 nodes each, 25 joint) and three types of equal share, loading $0.10\,\tilde{z}\tilde{b}$ on $y$, on $y^{(2)}$, and $0.071\,\tilde{z}\tilde{b}$ on both.

- `vy9t2`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; a firm's type is a pair $(f_1, f_2)$ drawn independently from $\{0.02, 0.07, 0.14\}$ for each state, nine types of equal share, and its cash flows load $e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each state.

**Verdict.** More factors are a lever only where the second state brings a second interaction. In BGN the second chain raises room by a quarter and doubles the gap (+0.015, ten of ten seeds). In GS21 it raises the product economy's gap from +0.020 to +0.056 (nineteen of twenty seeds), as the two-type version did (appendix). In KP14 it does not help however it is wired: the nested nine-type cell lowers room from 24% to 15% of the linear ceiling and the gap from +0.015 to +0.005; the three-direction version (`vym2t3`) keeps the room (28%) but halves the gap (+0.008); pricing each state at the full 2.5 (`vy9t2f`) only returns the gap to 3b's (+0.015, nine of ten seeds) at lower room (19%). The reason is in the ceilings: with an independent second type a firm's total exposure is the sum of two ladder draws, six evenly spread values that size and momentum proxy almost linearly, so the linear ceiling rises more than the nonlinear one (0.346 to 0.387). Both states act through the same channel, the growth-option elasticity, so the second state adds premium and a factor but not a second interaction. BGN's nine-type version (`bvy9t2`, +0.012, room 16%) sits between 3b and the three-direction cell. The reduced form (Table 7) says the same with the structure controlled.

### Reduced form: bounded and squared loadings, the sign ladder and the shifted squares

The RF rows of Table 8 are the `kfh` ladder with bounded (tanh) and squared loadings on a random index of the characteristics, the same squares with the size coordinate removed (`kfo`), the sign ladder, and the shifted squares of one state; the squares of one state at mean zero are in Table 7.

Bounded loadings behave like linear ones: no room at $K = 1$ and 4, a little at $K = 16$ (12%), and a negative gap throughout, because tanh of a single index is monotone in that index and nearly linear in its rank. The square of a random index gives room of about five times the linear ceiling and a gap of about +0.5 at $K = 1$; in the unconstrained square rows the gap falls to +0.12 and +0.14 at $K = 4$ and 16 because the squared index contains size$^2$, which the value-weighted Fama–French legs hold; the `kfo` rows remove the size coordinate and the gap returns to about +0.5 at every $K$.

The sign ladder (`lsh` rows) shifts the mean of one or both of the two latent factors in the $K = 1$ product by 0, 1, 2 or 3 standard deviations, and `lrt` replaces the product by a one-signed ratio of the two (a debt-service coverage ratio with a ridge). The mean-zero product has room of +0.39 (no noise) and +0.42 (noise 1.0) and fair gaps of +0.031 and +0.093, the $K = 1$ latent-product cell of the reduced-form table; every shifted cell and the ratio have room of 0.000 to 0.005 and a fair gap of −0.005 to −0.014, at the same $\mathrm{SR}_{\max}$ of 0.598. The linear-in-latents $R^2$ of the loading goes from 0.01 at no shift to 0.79 at a two-sd shift of one factor and 0.89 at two-sd shifts of both. Room in a one-factor economy is therefore all or nothing: a linear portfolio with any nonzero covariance with the exposure mimics the factor, diversifies the rest and takes its whole Sharpe, and only an exposure orthogonal to every linear function of the characteristics leaves the factor to the nonlinear estimator. This is what the structural sign-switching economies of Section 3b test.